Smart Poultry Farm Management
and
Disease Diagnosis

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Smart Poultry Farm Management and Disease Diagnosis

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DECLARATION

We solemnly declare that this report is written by us and is not copied from any online or printed material.

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List of Abbreviation

SVM: Support Vector Machine

CNN: Convolution Neural Network

ROC: Receiver Operating Characteristic

AI: Artificial Intelligence

DL: Deep Learning

GPU: Graphical Processor Unit

PDF: Probability Density Function

IoT: Internet of Things

ML: Machine Learning

Acknowledgement

We seek the guidance of the Almighty Allah to lead us in our lives and bestow us with His blessings and the wisdom to lead a righteous life. We also express our gratitude for the gift of education and knowledge, enabling us to contribute to the betterment of humanity. Without the support of Allah, we acknowledge that we would not have achieved anything. We are deeply thankful to our parents for their nurturing care from our earliest days. Furthermore, we are sincerely grateful to the Department of Electrical Engineering and Technology for their significant role in developing both our interpersonal and practical skills. We would like to express our heartfelt gratitude. We want to extend our special thanks to our supervisor, who provided us with the wonderful opportunity to work on the project "Smart Poultry Farm Management and Disease Diagnosis." Their guidance and support were invaluable as we delved into new areas of investigation. Additionally, we would like to acknowledge and thank our parents and friends who supported us in completing this project within a limited timeframe. First and foremost, we would like to convey our sincere appreciation to our teacher, Mr. Waseem Arshad, for his enthusiasm, valuable knowledge, and unwavering guidance. The valuable insights have significantly contributed to our investigation. He has been instrumental in shaping our report and has worked with us until late evenings. His extensive knowledge, remarkable experience, and expert skills have enabled us to successfully conclude this project. Without his support and guidance, we could not have envisioned a superior outcome in our thesis. He has been involved throughout our project planning and implementation. They were consistently available to provide guidance and technical advice whenever needed.

Abstract

As in the today's poultry farm, the proper monitoring, control and management of the system is not proper. The proposed system is monitored manually and managed by the labours. There is no proper monitoring of chickens according to the different life cycles of chickens at poultry farms. This leads to the improper management and control of the environmental factors according to the different phases of chicken growth cycle which increases the mortality rate and ultimately there are huge losses.

The goal of this project is to develop a system that can monitor, control and manage the environmental factors like carbon dioxide level, methane level, temperature and humidity. We can control the system automatically and manually through Mobile APP. To achieve the real time monitoring and control, we have to develop an APP for the manual and automatic control. We set the limits of temperatures, humidity and intensity of light in the environment according to the life cycle of hens.

To achieve this, a dataset of 1,020 images was collected from various sources, including the different Image Dataset, Keras, Kaggle dataset, Zonodo Dataset, GitHub also collecting from different poultry farms. These images were organized into four separate folders based on their respective conditions: coccidiosis, Salmonella, Newcastle, and normal. The size of every image in the dataset is adjusted to 128x128 pixels to ensure the uniformity. Preprocessing techniques, such as image resizing and filtering, were applied to enhance the image quality and reduce class imbalance.

The model was developed using a deep convolutional neural network (CNN) architecture. Different pre-trained models, such as MobileNetV2 was evaluated to find the best-performing one. Trained model was tested on the datasets. The performance of the model was evaluated using various techniques, such as accuracy, precision.

Once the model was trained and tested, it was deployed in an application to provide maximum flexibility and scalability. The user can upload an image of their chickens, and the app is detecting the image. The user can then take this information to their doctor for further examination and treatment.

The results achieved were highly promising, with an overall accuracy of 95% on the test set. The precision and recall for each class were also high, indicating that the model was able to detect the chicken diseases accurately. The model was also able to handle class imbalance, which is common in natural occurrence datasets like this one.

In conclusion, this project presents a smart monitoring, controlling and management of poultry farms with a deep learning-based approach to detect chicken diseases using images. The developed app provides a cheap, fast, and scalable solution for the monitoring and controlling of environmental factors according to the growth of the chickens at poultry farms and early detection of chicken diseases, which is necessary to prevent death. The model achieved high accuracy and performed well on a variety of evaluation. It can be deployed in mobile, and desktop applications to provide maximum flexibility and scalability. Overall, the project successfully achieved its objectives and has the potential to improve the millions of poultry farms suffering from these problems.

CHAPTER 1: INTRODUCTION

1.1 Background

There is lack of automated poultry farms at the recent times. All the work about monitoring and management is manual. We have no such developed systems where there is no need of labours for the feeding of chickens at poultry farms, controlling of environmental factors and detect the diseases earlier that brings the system more suitable for the chickens at poultry farms. The proposed system sets the parameters for the chickens according the different phases of chicken.

The system running by the workers at poultry farms that is not considers as the best, efficient and accurate method to control, manages and monitor the different factors according to the environmental conditions. The main issue of the manual controlling and monitoring is that there will be a delay in temperature sensing, humidity sensing, different dangerous gases like carbon monoxide, carbon dioxide, methane gas, ammonia gas sensing, water level monitoring, feed on and off monitoring, lighting system monitoring. This delay caused the dangerous effects on the growth of chicken at poultry farms and the life of chickens become miserable.

In recent times, there’s been a considerable increase in the utilization of machine learning (ML) techniques in the field of medicine. ML-based diagnosis of chicken diseases such as coccidiosis, Newcastle, and Salmonella has shown promising results in detecting and classifying these diseases accurately and efficiently.

Animals' digestive tracts can become infected with the parasite disease coccidiosis. Coccidia is the protozoan that causes it. The respiratory, neurological, and digestive systems of birds are impacted by Newcastle disease (ND), a highly contagious and deadly viral illness. Chickens are susceptible to a bacterial infection called salmonella. In chickens, especially young ones, it can result in systemic illness. To overcome these limitations, machine learning techniques have been applied to develop automated systems for diagnosing chicken diseases. These systems use different types of data, including images, to detect and classify diseases accurately and efficiently.

A vital part of the agricultural sector, the chicken business makes a substantial contribution to both economic stability and global food security. Effective and sustainable poultry farming methods are becoming more and more necessary as the demand for poultry products rises. Conventional poultry farming techniques frequently depend on manual administration and monitoring, which can result in inefficiencies, higher labour expenses, and a delay in responding to environmental and health concerns in poultry farms. Furthermore, disease outbreaks continue to pose a hazard, affecting food safety and resulting in large financial losses.

One potential approach to addressing these issues is to include smart technologies into the management of chicken farms. To automate and optimize farm operations, smart poultry farm management makes use of sensor networks, data analytics, Internet of Things (IoT) devices, and artificial intelligence (AI). These technologies allow automated control of feeding, lighting, and ventilation systems, as well as real-time monitoring of environmental parameters including temperature, humidity, and air quality. AI-driven disease diagnosis systems also use data analysis, machine learning algorithms, and image processing to identify early disease indicators in poultry, allowing for timely intervention and lowering the possibility of extensive epidemics. The goal of this project is to create a comprehensive smart poultry farm management and disease diagnosis system that will reduce disease-related losses while simultaneously increasing farm productivity and efficiency.

Our project aims to develop a smart control, management and monitoring of the environmental factors along with the machine learning-based system for diagnosing coccidiosis, Newcastle, and Salmonella using images captured by a mobile camera and uploading images from gallery. The system is using deep learning algorithms to analyze the chicken images and classify them into different disease categories. We have carefully selected an algorithm that has been shown to perform well in image classification tasks, making it an ideal choice for our project.

In conclusion, our project aims to develop a system with automatically set parameters according to the stages of chickens also with manual control through app and a machine learning-based system for diagnosing Coccidiosis, Newcastle, and Salmonella using images captured by a mobile camera or by uploading the images from the gallery. The system is running and is using deep learning algorithms to classify the images. The suggested strategy will help create an ecosystem for poultry farming that is robust, sustainable, and profitable by incorporating contemporary technologies. accurately and efficiently. Additionally, an app has been developed that is providing information about. The system's low cost and ease of deployment make it a promising solution for the real time monitoring, control, management and early detection and prevention of coccidiosis, Newcastle, and Salmonella in underserved areas.

1.2 Background of study

The most significant sector in the agricultural sector contributes to the economic growth and food security is poultry farming. Due to the increasing needs of poultry farm's products like meat and eggs, poultry farms are extended with more and more significant and efficient management system. As looking toward the poultry farms of today's society, it is dependent on the manual system for monitoring, controlling and management of the whole system, refilling the containers and maintain environmental conditions. So, by using this manual method, there is the inefficient system generated with inconsistent monitoring and due to this the disease diagnosis process is delayed, that brings disaster.

The major challenges of the traditional or manual system are maintain optimized environmental conditions like temperature, light intensity, humidity, level of carbon mono-oxide, etc. these environmental factors are very important for the growth of chickens at different phases of life. Manual monitoring for these factors leads to human error and inefficient monitoring and management that results to poor farm conditions, increased morality rates and decreased in production. Moreover, the improper feeding system leads to misuse of resources, increase in costs and imbalance of nutrition in diet of chickens.

To overcome these challenges, by using the smart technologies into poultry farm gives the solution. The use of different sensors like temperature sensors, humidity sensor and different environmental controlled parameters' sensors, automation, artificial intelligence gives the system with real time monitoring and managing of the environmental factors. It automates feeding, water level monitoring, lighting system to detect the disease early and resource optimization.

1.3 Problem Statement

The main problem for the poultry farms is deficiency of real time monitoring and management. Our main challenge is to make the poultry farms automated where all the factors related to the chickens are monitored by the sensors and make the system more efficient and make it for real time management. We have to set the parameters according to the phase and life cycles of the chickens like brooding and growing stages.

Coccidiosis, Newcastle, and Salmonella are major diseases that pose significant challenges to global health. These conditions affect millions of poultry farms worldwide, and if left untreated, they can have severe consequences on them and quality of life.
The accurate and timely diagnosis of these diseases, including Coccidiosis, Newcastle, and Salmonella, is crucial for effective treatment and preservation of life. However, traditional diagnostic methods for these diseases often require expert knowledge and can be time-consuming. Moreover, misdiagnosis or delayed diagnosis may result in improper treatment and potential loss. Majorly, real- time monitoring is very important at poultry farms. Sometimes, the diseases did not detect earlier so, it will ultimately cause of spreading the disease from a single chick to whole poultry farm. This limits their availability to healthcare professionals and specialized clinics.

Therefore, there is a need to develop an automated and efficient diagnostic system using machine learning techniques.

1.4 Objective

The main objective of the system is to make the system more efficient, efficient and sustainable. It is only possible by making the system real time monitoring and manages the data according to the need. The proposed system made the system for real time monitoring by using the different sensors and microcontroller for the accessibility of parameters that affects the chicken's life. Real time monitoring is done by the app development and deploy the ML model into the mobile app for the real time monitoring and detect the disease in the chickens earlier.

A machine learning-based system for the early detection and diagnosis of coccidiosis, Newcastle and Salmonella is what this study's goal is. The device employs a mobile camera to take an image of the feces, and running up deep learning algorithms interpret the image. An app that can be used to view the findings of the diagnostic is also being part of the system.

The objective of this study is to advance system of the poultry farms by offering a low-cost, non-invasive, and precise approach for the early identification and diagnosis of coccidiosis, Newcastle and Salmonella. Early identification is essential for preventing the chickens of poultry farms at worldwide level.

To achieve these objectives and aims, the study will include the following tasks:

- Real time management

  The real time monitoring is done by the usage of different sensors, integrated with the hardware.

- Microcontroller

  Esp32 microcontroller is to integrate the hardware with sensors.

- App Development

  We have to develop an App for monitoring the poultry farm's environmental parameters and to detect the diseases by taking the images from camera and by taking picture through camera.

- Development of Dataset:

  The creation of a dataset involves gathering and curating photos of feces in various states of health, including those with coccidiosis, Newcastle and Salmonella. The deep learning model is tested and trained using this dataset.

- Preprocessing of images:

  Preprocessing is done on the photos to get rid of noise and artefacts, boost contrast, and normalize intensity. We'll employ a variety of image processing methods, such as filtering, equalization, and morphological procedures.

- Development of Deep Learning Models:

  MoblieNetV2 is actually the Convolutional neural networks (CNNs) model that is able to run on mobile devices. It will be created and trained on the dataset. The models will be improved utilizing methods like transfer learning and hyperparameter tweaking.

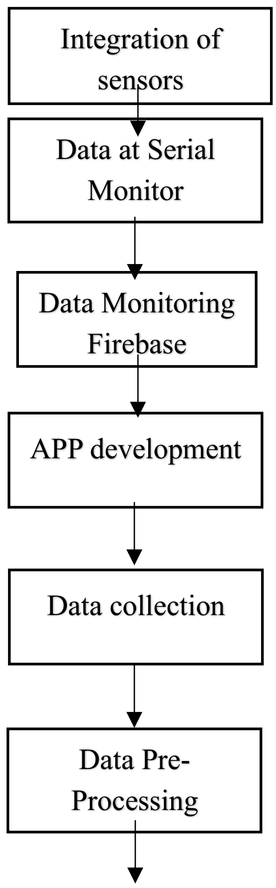
- Evaluation of Models:

  The developed models will be evaluated using various metrics, such as sensitivity, accuracy, specificity, sensitivity and area under the receiver operating characteristic (ROC) curve. The models will also be compared to existing methods for diagnosis of Coccidiosis, Newcastle and Salmonella.

- Implementation of system:

  The developed system will be implemented and integrated with a Mobile camera. The system will also include an app that can be used to view the results of the diagnosis.

By offering a low-cost, non-invasive, and precise tool for the early identification and diagnosis of coccidiosis, Newcastle and Salmonella. The suggested system's flowchart is shown in the figure below.



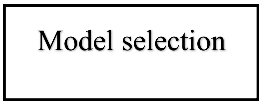


Figure 1 Flowchart of Proposed system 1

1.5 Scope of Study

The main focus of the system is to monitor, control and manages the environment of the poultry farm. The parameters like temperature, humidity, light intensity are controlled and monitored by the App manually and automatically. The second main purpose of the project is to detect the chicken diseases earlier.

Our poultry farms are not enough up to date that it is not controlled and monitored by the smart technologies and terminologies. It is major cause of the chicken that leads to death and owner faces difficult situation.

Coccidiosis, Newcastle and Salmonella are some of the most common chicken diseases globally, responsible for a significant number of cases worldwide. Early detection and treatment of these diseases are critical for preserving and preventing chickens from these diseases. However, manual diagnosis by professionals can be time-consuming and subjective, and lack of access to facilities is a significant challenge, especially in developing countries.

To address these challenges, this study aims to develop an automated system using machine learning for the diagnosis of Coccidiosis, Newcastle and Salmonella. The system utilizes a mobile camera to capture images of the feces of infected chickens, which are then processed using code. The processed images are used to diagnose the presence of Coccidiosis, Newcastle and Salmonella. Additionally, an app will be developed to inform the user of the diagnosis and provide information about the disease.

The use of ML based systems can provide quick and objective results, enabling early detection and treatment of Coccidiosis, Newcastle and Salmonella diseases. This study's system utilizes a mobile camera, making it accessible and affordable, especially in low-resource settings where traditional facilities may not be available. Our updated system provides a low-cost and portable platform for processing the images, making the system practical for use in the field.

The development of the app is a significant addition to the potential impact of this study. The app can help to educate and empower individuals who may not have access to traditional care services by providing information about the disease and diagnosis. This can lead to increased awareness and early detection of diseases, ultimately leading to better outcomes.

The study's focus on the development of the APP for manual and automatic control, uses firebase platform for this purpose and also use of deep learning techniques provides an opportunity to explore the potential of these networks in the diagnosis of diseases. By comparing the performance of different deep learning models, this study can provide insights into the most effective methods for diagnosing Coccidiosis, Newcastle and Salmonella using machine learning.

The development of an automated system for controlling, managing and monitoring of the whole environment, the diagnosis of Coccidiosis, Newcastle and Salmonella using machine learning is a significant step towards early detection and treatment of these diseases. The use of a mobile camera and upload the image from the gallery makes the system accessible and affordable, particularly in low-resource settings. The app's development further adds to the potential impact of this study, promoting awareness and early detection of diseases. The study's focus on the use of deep learning techniques provides insights into the most effective methods for diagnosing these diseases using machine learning.

1.6 Application Areas

Smart poultry farm management and disease diagnosis systems have utilized in several areas within poultry farms, ensures high production rate, reduces mortality rate, having managed resources and better disease diagnosed.

The application areas of this system are commercial poultry farms, small and medium poultry farms, breeding farms, research and development in agriculture, smart farming, sustainable farming, government departments, AI analysis for poultry farms, poultry farm automation and remote monitoring.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction to modern management system

The growing demand of industry that provides eggs and meat to fulfil the global demand is poultry farm industry. The manual method faces many problems like inefficient resources usage, increasing costs for labour, and late detection of diseases. To overcome these problem, modern poultry farms are used where automation, Internet of Things and artificial intelligence embedded. By using these techniques, imbalanced diet of chickens is revolutionised to balanced diet of chickens, increased the production rate and early detection of disease.

Farmers can increase the efficiency of the system by using this modern management system.

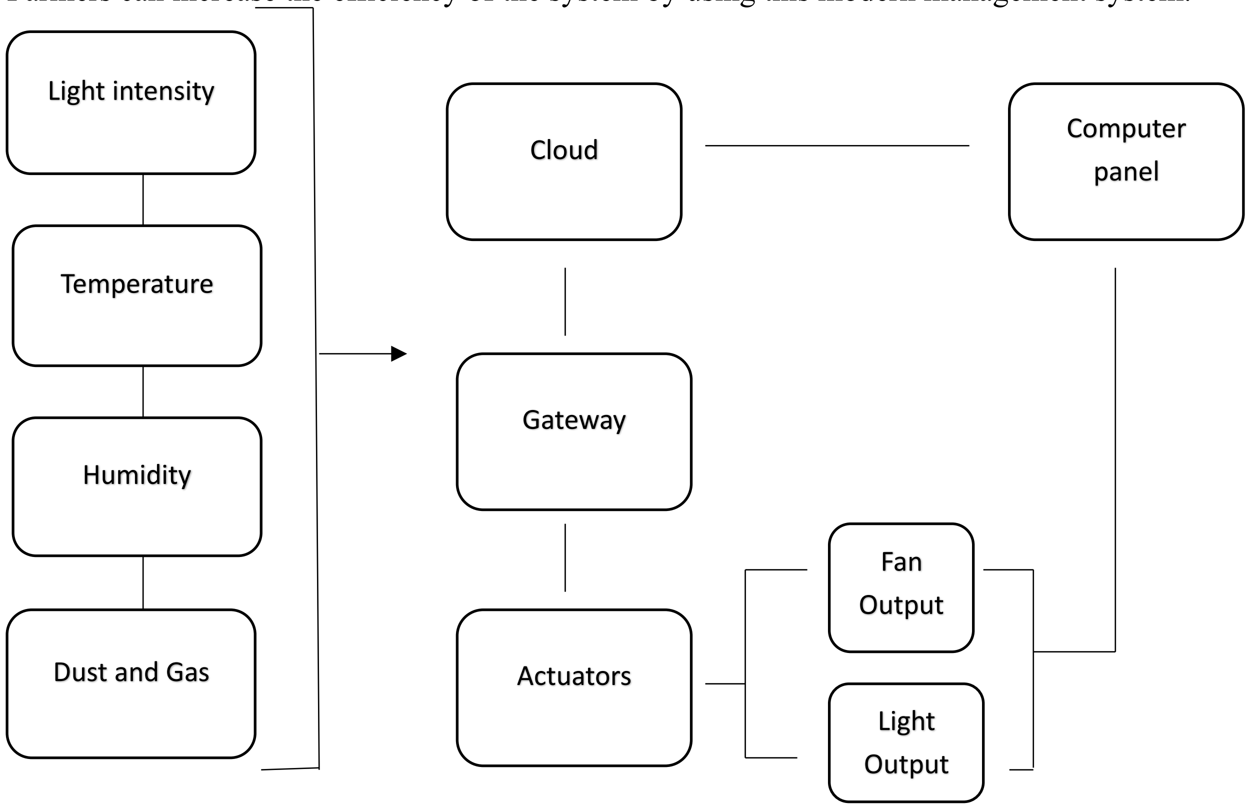


Figure2 Proposed system of management 1

2.2 Life cycle of Chicken Growth

2.2.1 Brooding Cycle

The period after the chicken hatch is brooding cycle. It requires external heat, special care and external nutrition before at this stage of chickens, they cannot be able to regulate their body temperature.

The total days for brooding cycle are ten days where the temperature requirement for the this cycle is from 25 degrees Celsius to 30 degrees Celsius.

2.2.2 Growing cycle

The growing cycle comes after the brooding cycle and comes before the maturity level. It is the cycle where the chickens have to develop into either egg laying eggs or meat producing birds. It is very important stage for the weight gain, development of organs and resistance of diseases. The temperature ranges required for the growing cycle is from 28 degrees Celsius to 35 degrees Celsius. This cycle is almost fourteen days long.

2.3 Role of Automation at poultry farms

Internet of things (IoT) is one of the emerging techniques that is used to automized the whole systems using different sensors. Manual system is converted into smart systems by using these services. At poultry farms, different sensors are used to made the manual poultry farm systems to smart poultry farm system. At poultry farms, farms can use sensors like temperature sensor, humidity sensor, light sensor, methane level sensor, carbon mono-oxide level check sensor, air quality sensors to make the best management system to bring the results more than average. Automated feeding and automated water management system is used in poultry farms to make the system for real time monitoring.

2.3.1 Different Sensors for Automation

2.3.1.1 MQ-135

Different gases that present in environment like ammonia gas, Benzene, Sulphur gas, etc are detected by a special sensor named as MQ-135. These gases affect the whole environment badly. It has 4 pins named as GND, input voltage VCC, analog output pin and digital output pin. This sensor operates at given operating output voltage given by any microcontroller.

2.3.1.2 MQ 4

MQ4 is the air quality sensor that senses the methane gas in the environment. It also uses at the homes to detect the leakages of methane and CNG gas. The pin configuration of the senor is as it consists of four pins for operating it efficiently. MQ4 has 4 pins such as VCC pin for input operating voltage, GND pin, Analog pin for analog output and digital pin for digital output.

2.3.1.3 MQ 7

The air quality sensor that detects the carbon monoxide level in the environment is called MQ7 sensor [1]. It senses the concentration of CO in the air [2]. It has senses the different gases very efficiently and gives real time response. It consists of four pins; VCC pin, GND pin, Aout (analog output pin), Dout (Digital output pin).

2.3.1.4 DHT11

DHT11 is the sensor that detect temperature and humidity from the environment. 50 degrees Celsius is the maximum temperature detected by the sensor. [3] It consists of three pins such as VCC for input operating voltage, GND for ground pin and Data pin for receiving output signal.

2.3.1.5 Aj-SR04M

Aj-SR04M is the ultrasonic sensor that is used to measure hight or distance from certain point. It is mainly used for outside application where the environmental condition is hard and moisture is present. It has four pins such as GND, ECO, TRIGGER, 5V DC.

2.3.2 Microcontroller

2.3.2.1 ESP 32

In Esp32, different wireless communications done along with the low cost and highly efficient. It cooperates essential components for wireless communication such antenna switches. It provides connection of different types of communication with many devices for wireless communication.

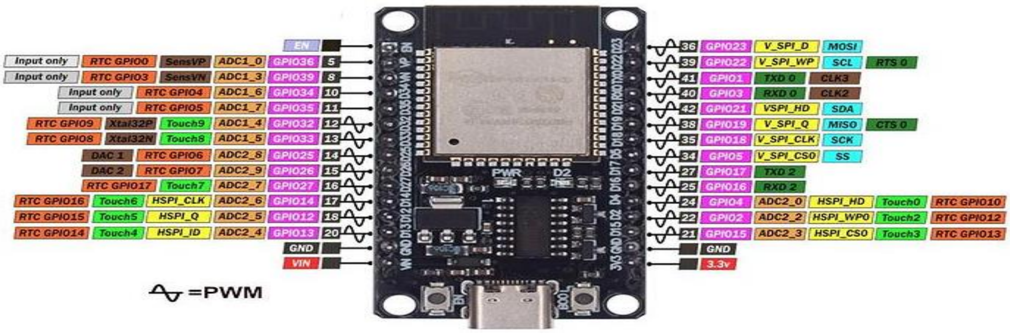


Figure3 Datasheet ESP32 microcontroller 1

2.3.3 Firebase Platform

Google developed a backend- as-a-service platform named as Firebase [4]. It provides us different tools and several services. It helps to develop the web and mobile development. The main features of firebase are real-time database, cloud storage, cloud massaging, cloud functions, hosting, etc.

2.4 Current diagnosis method and limitations

The current procedure for diagnosis of coccidiosis, Newcastle and Salmonella is from testing the feces of infected chicken. For coccidiosis, the examine for the feces is taken, sometime the sample under the microscope is not precise, that's why the delayed detection and variable oocyst shedding occurs. For Newcastle, the laboratory test at the initial stage is very important. The virus can be isolated from tissues from infected birds. For Salmonella, from the infected chicken, the sample of droppings and blood is taken. The laboratory test detects the salmonella bacteria in the stool or blood of infected chicken. However, it is time taken process, expensive, and require specialized equipment and expertise. In addition, human error is the main risk for this process or bias that can affect the interpretation of the results. Furthermore, the accuracy of the tests may vary depending on the skill and experience of the examiner, leading to inconsistent diagnosis and treatment.

Machine learning-based diagnostic methods, on the other hand, offer the potential for faster, more objective, and accurate diagnoses of these conditions. These methods can analyze large amounts of data from different sources, including images captured by mobile cameras, and provide automated and consistent diagnosis with high accuracy. However, there are a few limitations to these methods.

The primary challenges are the requirement for large amounts of high-quality training data to develop and validate the models. Additionally, the interpretability of these models may hinder their acceptance in practice. Despite these limitations, researchers are exploring various techniques to address these issues.

Transfer learning is a technique that enables models to learn from pre-trained models and adapt to new data with limited training data. Data augmentation techniques can generate synthetic data to increase the diversity and size of the training data, thereby improving the performance of the models [5]. Explainable AI techniques provide insights into how the model makes its predictions, increasing the transparency and trust of the model.

While tradition diagnosis methods for Coccidiosis, Newcastle, and Salmonella have their limitations, machine learning based diagnostic method offer potential benefits in term of accuracy, consistency and speed. The application of techniques such as transfer learning, data augmentation and explainable AI can further enhance the accuracy and reliability of these models. However, it is important to ensure that these models validated and proven effective before they can be integrated into clinical practice.

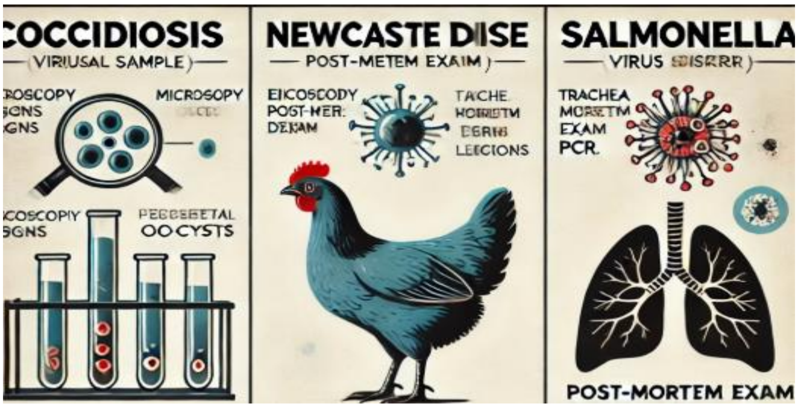


Figure4 laboratory test in lab 1

2.5 Introduction to Machine Learning in Disease diagnosis

Medical diagnosis has been highly improved with the use of machine learning, specifically in the field where various techniques have been applied for the diagnosis and detection of diseases such as Coccidiosis, Newcastle and Salmonella. These diseases are among the leading causes worldwide. Specialists have been using laboratorial techniques like testing to diagnose these diseases, and machine learning has been successful in aiding them.

One of the most commonly used techniques for testing classification is deep learning. This involves using pre-trained neural networks to fine-tune on a new dataset. This technique has shown promising results in detecting and diagnosing Coccidiosis, Newcastle and Salmonella. In addition, support vector machines and decision trees have been utilized for classification of images in the context of diagnosis.

However, there are still some challenges that need to be addressed in applying machine learning techniques to disease diagnosis. One challenge is the limited availability of labeled datasets, which makes it difficult for machine learning algorithms to train and generalize.

Another challenge is the need for interpretable models that can provide insights into the disease diagnosis process. Furthermore, robust models that can handle noise and variations in the data are necessary.

In the future, deep learning models like generative adversarial networks and variational auto encoders may be used for disease diagnosis. These models have shown potential in generating synthetic images that can augment the limited datasets available for diagnosis. Additionally, interpretable machine learning techniques such as decision trees and rule-based models may be developed to provide insights into the disease diagnosis process.

Machine learning has proven to be a useful tool in disease diagnosis, particularly in this field where it has been applied for the detection and diagnosis of Coccidiosis, Newcastle and Salmonella diseases. While there are still challenges that need to be addressed, the potential of machine learning in improving the accuracy and efficiency of disease detection is promising. As technology continues to advance, it is likely that machine learning will play an even greater role in diagnosis in the future.

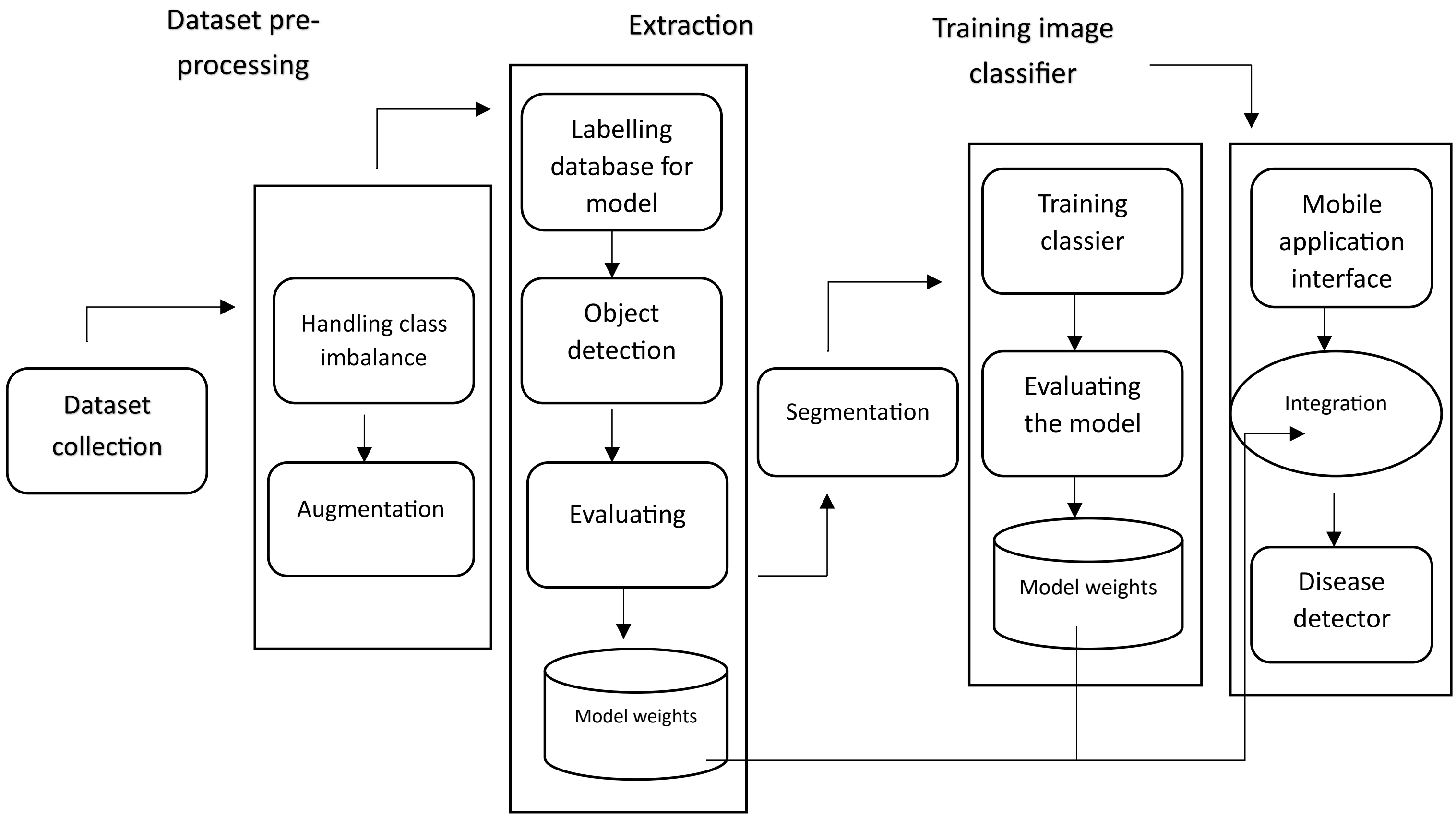


Figure5 Disease Diagnosis through DL 1

2.6 Poultry diseases: Coccidiosis, Newcastle, and Salmonella

Coccidiosis, Newcastle, and Salmonella are the most common Chicken diseases affecting millions of chickens worldwide at poultry farms. These diseases can lead to disaster and ultimate death of chickens at poultry farm if not detected and treated early. In this section, we will discuss these three diseases in detail, including their causes, symptoms, and available treatments.

2.6.1 Coccidiosis

Animals with coccidiosis suffer from a parasitic intestinal ailment brought on by coccidian parasites. Animals can contract the disease by consuming diseased tissue or coming into touch with infected faeces. The main symptom is diarrhea, which can turn bloody in extreme situations. While the majority of coccidia-infected animals show no signs, young or immunocompromised animals may experience severe symptoms or even die.

The application of machine learning algorithms may aid in the early detection and accurate diagnosis of coccidiosis, resulting in timely medical intervention and improved treatment outcomes. Early diagnosis is crucial as it can delay the progression of coccidiosis and alleviate discomfort. With advancements in technology, machine learning algorithms can analyze medical imaging data and identify patterns indicative of coccidiosis. These algorithms can then provide a more comprehensive analysis of the images and assist healthcare professionals in making more informed decisions regarding treatment.

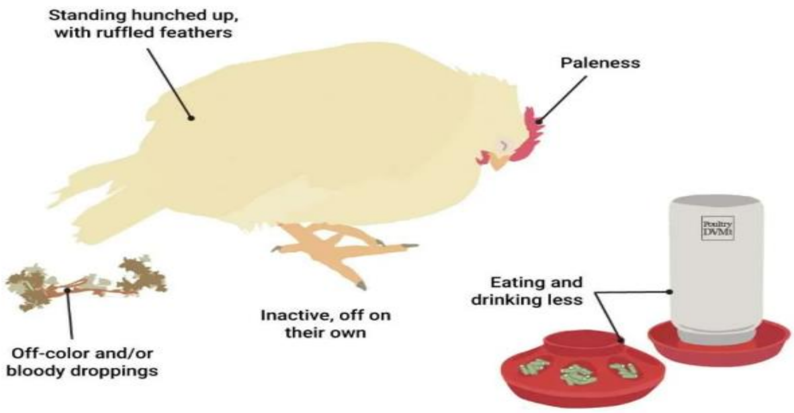


Figure6 Way to Diagnose Coccidiosis 1

2.6.2 Newcastle

Numerous domestic and wild bird species are susceptible to Newcastle Disease, a highly contagious viral infection, to varied degrees. Clinical signs of the disease may include digestive, respiratory, and/or neurological symptoms. These symptoms can range from a mild, nearly undetectable respiratory condition to very severe depression, decreased egg production, increased respiration, diarrhea that is followed by collapse, or, if the birds survive, long-term nervous symptoms (like twisted necks). The disease is extremely deadly in its most severe stages. The paramyxovirus that causes Newcastle disease can range in virulence from mild to highly virulent. Direct physical contact with diseased or infected birds is typically how the disease is spread. Manure contains the virus, which is inhaled into the atmosphere [6].

Contaminated clothing, food, drink, carcasses, and equipment are other sources of infection. The virus is easily spread by people from one farm or shed to another. Although humans can get conjunctivitis from the Newcastle Disease virus, humans are not as susceptible as birds are.

Figure7 Chicken with Newcastle Disease 1

2.6.3 Salmonella

Salmonella enterica species, including S. Gallinarum, S. Pullorum, S. Enteritidis, and S. Typhimurium, are the main culprits behind Salmonella, a serious bacterial disease that affects poultry. It poses significant risks to poultry health and human food safety through zoonotic transmission. The disease is transferred horizontally through contaminated feed, water, equipment, and contact with diseased birds, and vertically from breeders to eggs. It can also spread by environmental vectors like farm workers, rodents, and wild birds.

The age and strain of the hens affect the clinical symptoms. While fowl typhoid causes depression, pale combs, decreased egg production, and occasionally greenish-yellow diarrhea in adult birds, pullorum illness causes white diarrhea, lethargy, and high mortality in chicks. In addition to raising concerns about food safety because of tainted poultry products, the disease results in significant mortality, decreased production, and higher control costs, all of which leads to financial losses.

Vaccination, hygiene management, biosecurity measures, and routine health monitoring are all necessary for effective control. By utilizing IoT devices, sensors, and AI-driven diagnostics to facilitate early detection, automate biosecurity procedures, and optimize farm management, smart poultry farm management systems can improve disease control, thereby lessening the impact of Salmonella and increasing production.

Figure8 Dropping of Healthy chicken 1

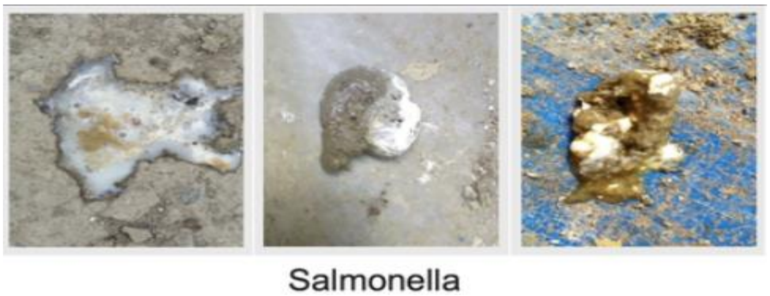


Figure9 dropping having Salmonella 1

2.7 Other Poultry Farm Diseases:

Coccidiosis, Newcastle, and Salmonella are the most common diseases that affects the chickens directly. There are many other poultry farm diseases that causes the chicken to ultimate death and affects our economy severely.

Marek's disease is very dangerous viral disease that causes the tumors and paralysis of chickens.

Bronchitis infection is also a viral disease that spread due to a special type of virus, the affected chicken is suffering from cough, sneeze and low production rate of eggs.

Mycoplasmosis is the bacterial disease that causes respiratory problems in chicken. Poor growth and low production rate are the major symptoms shown by the infected chickens.

Pullorum is the bacterial disease that affects the young chickens mostly. The young chickens are unable to survive with this severe condition of the disease. This disease is caused by the Salmonella pullorum bacteria and leaves high death rate [7].

2.8 Flaws of existing systems

The main flaw of the existing literature is; there is no real time monitoring of the system and lack of real time disease detection. Manual systems did not use any sensors for the real time monitoring. Existing systems are dependent on the manual system for monitoring the whole system, refilling the containers and maintain environmental conditions. So, by using this manual method, there is the inefficient system generated with inconsistent monitoring and due to this the disease diagnosis process is delayed, that brings disaster and increase mortality rate of chickens.

CHAPTER 3: METHODS AND MATERIALS

Our system strives to develop the smart management system where the environmental parameters are in control by using different sensors like temperature sensor, humidity sensor, air quality sensor, ammonia check sensor, carbon mono-oxide level check sensor, etc for real time monitoring. Also, our system aims to develop a machine learning based system for the real time diseases detection and diagnosis including Coccidiosis, Newcastle, and Salmonella using images captured by mobile camera. We made the TF lite file and deploy on Mobile app. We add the options such that uploading of the picture from gallery of the mobile and using mobile camera for taking pictures of chickens in mobile app. Additionally, we developed a mobile app that is also detecting the disease.

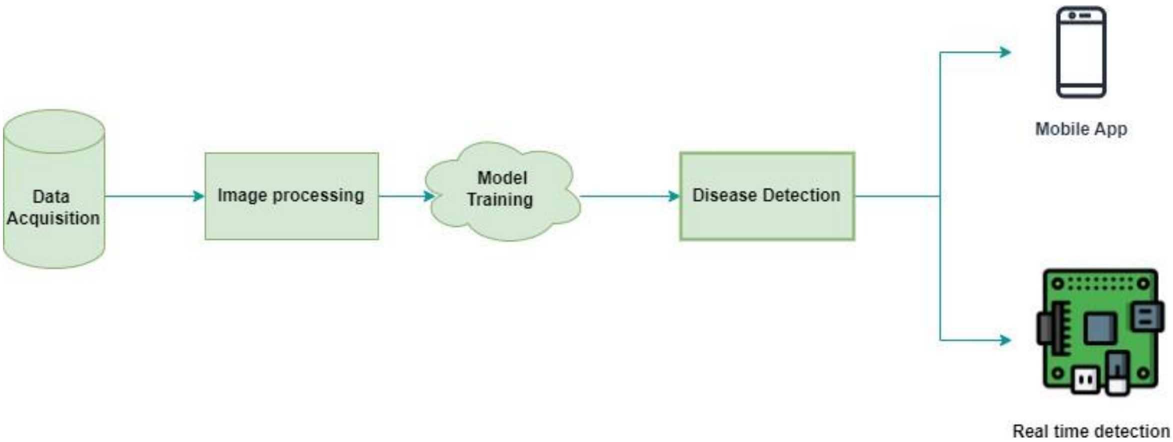


Figure10 Flowchart of Project Methods 1

3.1 Integration of Sensors:

Integration of sensors are very important for the automation of whole system. First of all, code of all the sensors are integrated to microcontroller. In our case, DHT11 sensor, MQ4 sensor, MQ7 sensor, MQ 135 sensor, Aj-SR04M sensor are integrated with ESP32 microcontroller. The data of all sensors are uploaded to the microcontroller ESP32. After uploading the code, the compiler compiles the overall code through WiFi. After establishing the WiFi connection, Data of all sensors appears on the serial monitor and make a link with Firebase.

3.1.1 Data on Serial Monitor

We can easily access data from the serial monitor of the compiler for the further connection. Following is the picture that shows the data of the different sensors

```
Connecting to Wi-Fi..
Connected to Wi-Fi!
IP Address: 192.168.100.11
Firebase setup successful
Cycle: Brooding | Temp: 23.4 °C | Humidity: 63.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 19 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 19 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 19 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 19 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 21 % | Methane: 25 % | CO: 18 %
Cycle: Brooding | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
Cycle: Growing | Temp: 23.8 °C | Humidity: 62.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
Cycle: Growing | Temp: 23.8 °C | Humidity: 61.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
Cycle: Growing | Temp: 23.8 °C | Humidity: 61.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
Cycle: Growing | Temp: 23.8 °C | Humidity: 61.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
Cycle: Growing | Temp: 23.8 °C | Humidity: 61.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
Cycle: Growing | Temp: 23.8 °C | Humidity: 61.0 % | Distance: 597 cm | Air Quality: 20 % | Methane: 25 % | CO: 18 %
```

Figure11 Data on Serial Monitor 1

3.1.2 Firebase Interfacing

Our code established a link to Firebase account using WiFi. Following is the picture that describes the interfacing of Firebase through WiFi.

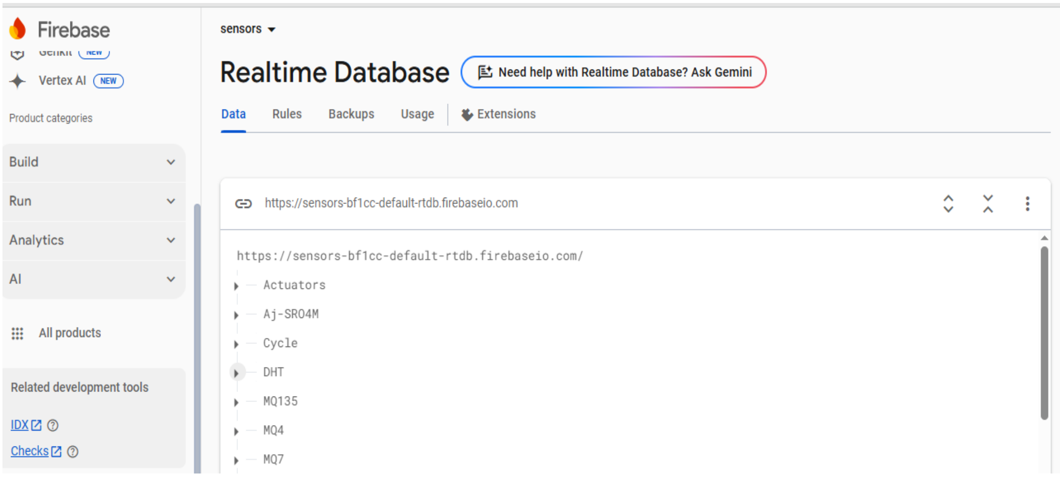


Figure12 Data of Sensors on Firebase 1

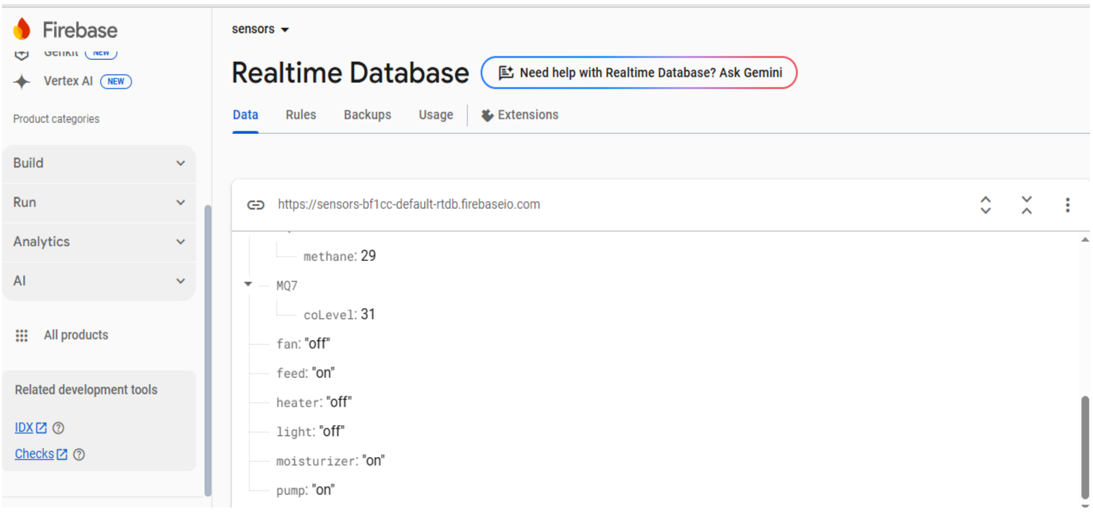


Figure13 Data of outputs on firebase 1

3.1.3 Integration of sensors with Hardware

Integration of sensors with hardware is very important part for the real time monitoring of data of poultry farm.

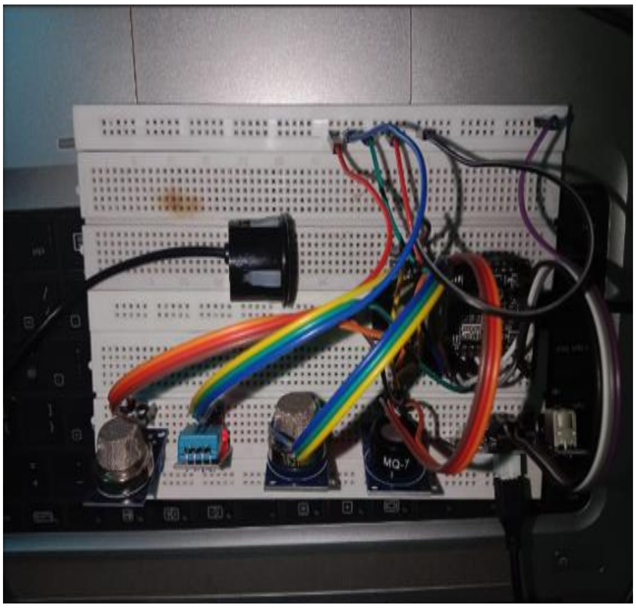


Figure14 Integration on Breadboard 1

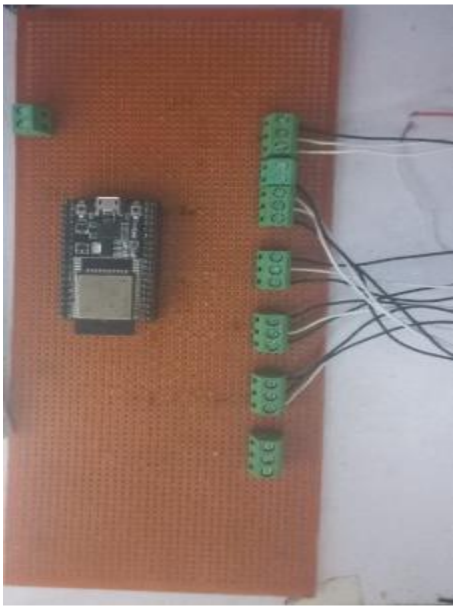


Figure15 Integration on Verroboard 1

3.1.4 Hardware Implementation

Integration of Sensors and different output things like Moisturizer pump, Light, Water pump, Heater and fan.

Figure16 Sensor integration 1

Figure17 Hardware prototype 1

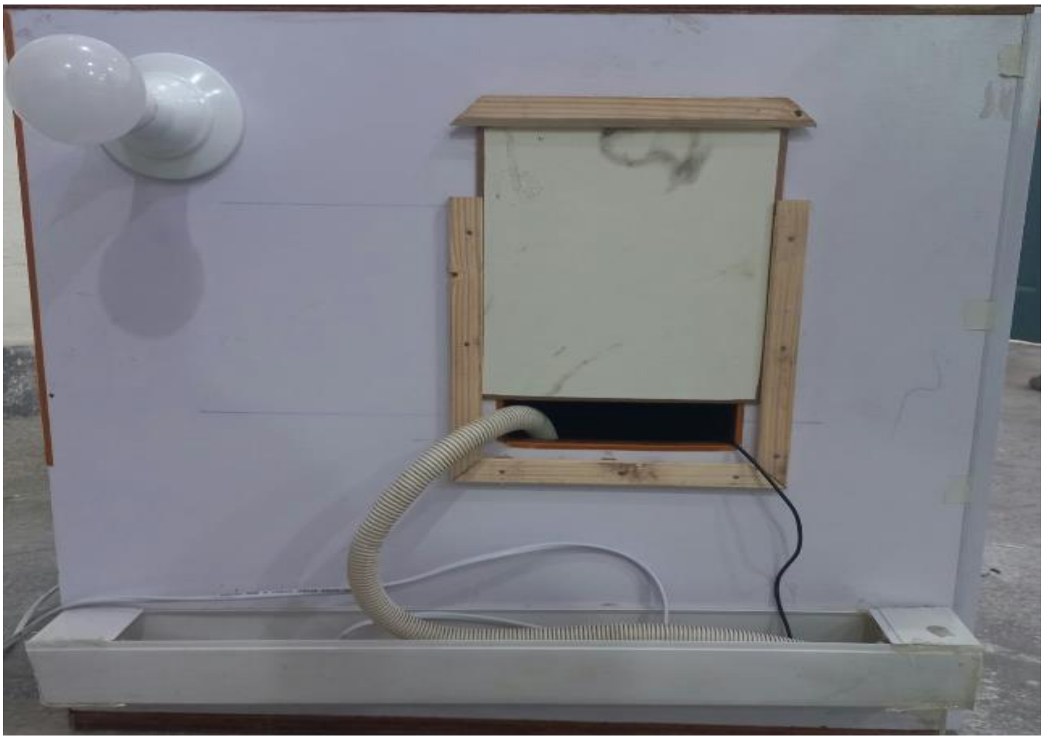


Figure18 Hardware Display 1

3.2 Data Acquisition

3.2.1 Image Datasets for chicken Diseases

Image datasets play a crucial role in the development of machine learning algorithms for the diagnosis of chicken diseases such as Coccidiosis, Newcastle, and Salmonella. In this field, the use of image datasets for diagnostic purposes has revolutionized the way to detect and cure the chicken diseases. The availability of large-scale image datasets has enabled the development of accurate and reliable machine learning models for the detection and classification of various chicken diseases.

In this chapter, we will discuss the different types of image datasets that have been used in the diagnosis of chicken diseases. We will also highlight the importance of using high-quality image datasets that are representative of real-world scenarios to ensure the accuracy and generalizability of machine learning models.

3.2.2 Types of Image Datasets

There are different types of image datasets that have been used in the diagnosis of chicken diseases. The most commonly used datasets are chicken disease image datasets, which consist of images of the back of the captured using a mobile camera. However, in our project, we are using a real chicken image dataset that consists of images captured using a mobile camera.

The use of real chicken disease image datasets is important as it provides a more accurate representation of the poultry farm. Real chicken disease images can capture a wide range of chicken conditions, including those that may not be visible in fundus images. Additionally, real images can capture the dynamic changes that occur in the, such as changes in size and movements, which cannot be captured by fundus images.

3.2.3 Sources of Image Datasets

For our project, we gathered and utilized various datasets to train our machine learning algorithm. The dataset we used for our project are taken from following sources:

3.2.3.1 Kaggle

The first dataset we used is from Kaggle. Kaggle is a platform where you can find various datasets, including medical datasets related to chicken diseases. We used several chicken disease datasets from Kaggle in our project.

Figure19 images for Salmonella 1

3.2.3.2 Keras

The second dataset we used is from Keras. Keras is a platform where you can find various datasets, including medical datasets related to chicken diseases. We used several chicken disease datasets from Keras in our project.

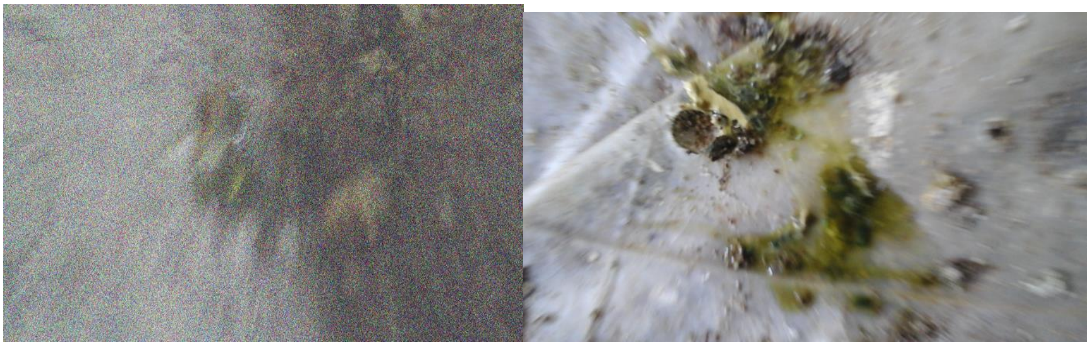


Figure20 images of Newcastle 1

3.2.3.3 Zonodo

It is free online platform for collecting images from it to use it as it is in the project. The third dataset we used is from Zonodo. Zonodo is a platform where you can find various datasets, including medical datasets related to chicken diseases. We used several chicken disease datasets from Zonodo in our project.

Figure21 Images for Coccidiosis 1

3.2.3.4 Dataset collected from several Poultry farms

Apart from these sources, we also gathered some datasets from poultry farms during our poultry farm visits. We made sure to use a diverse set of datasets to train our algorithm so that it could accurately detect various chicken diseases, including Coccidiosis, Newcastle, and Salmonella. By utilizing multiple sources, we aimed to ensure that our model would be robust and effective in real-world scenarios.

Figure22 Image of healthy chicken faeces 1

3.3 Image Processing

3.3.1 Overview

Data analysis, machine learning, and natural language processing (NLP) all require pre-processing as a critical step. It alludes to the set of procedures carried out on unprocessed data to get it ready for additional analysis. The imaging sensor, transmission channels, and the method of acquiring the image are just a few examples of the origins of image noise.The fundamental definition of image processing methods used to get rid of or cut down on this noise to raise the image's quality.

Over the previous ten years, a number of methods, including filtering, restoration, and enhancement techniques, have been created to improve image degradation. In order to provide a better and more accurate representation of the original image, these techniques are employed to eliminate noise, blur, or other distortions from the image.

A further explanation is Image processing involves applying mathematical techniques and algorithms to manipulate and analyse digital images [8]. It entails the application of a variety of computational techniques to improve the quality of a digital image, extract pertinent data, and prepare it for future examination or interpretation.

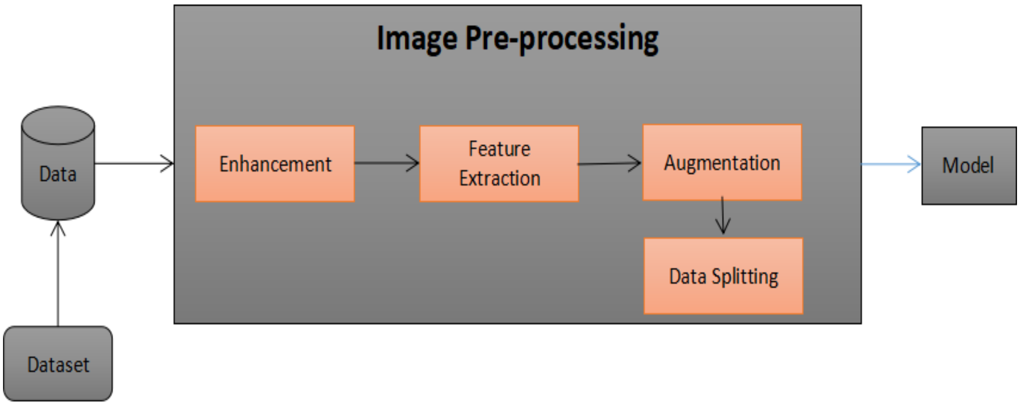


Figure23 Block diagram of Pre-processing 1

As mentioned in previous chapter, we are not using the already generated data set. Because our project dealing with the real life detection of chicken diseases. So that we generated the data set artificially. In deep learning model we need large data set so that we apply the data Augmentation on the artificially generated data set and enhance our data set.

3.3.2 Image Enhancement

Image enhancement is a term used to describe a group of methods used to increase the quality of an image prior to additional processing or analysis. Pre-processing image enhancement aims to fix any distortions or flaws in the image and improve its suitability for a certain application. It include changing an image's brightness, contrast, and noise levels as well as smoothing or sharpening the image and fixing aberrations like geometric or radiometric ones.

Overall, image enhancement in pre-processing is an important step in preparing an image for further analysis or processing, and it can greatly enhance the quality and appropriateness of the image for a certain application.

3.3.3 Feature Extraction

In above section, we discussed the long detail on the Image enhancement that is used to improve the image quality be reducing its noise and improving its brightness and contrast. Now we take a look to the next step of Pre-processing that is the Feature extraction. In feature extraction we remove the irrelevant information which helps to improve the accuracy and efficiency of the model.

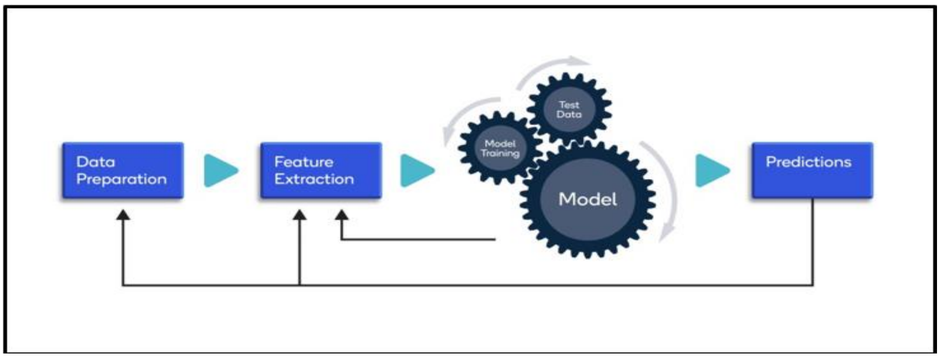


Figure24 Basic block diagram of ML 1

we prepare the data properly using different techniques. In data preparation we collect the data set and perform pre-processing using different techniques. After that we perform the Feature Extraction. In feature extraction we reduce the multiple features into the minimum features and these features give us the required information according to our scenario. Feature extraction is useful when you have a large number of dataset and you want to reduce without losing any important and relevant information [9]. Feature extraction is used to reduce the amount of redundant data from the dataset. In the end, less machine efforts required to build the model and also increase the speed of learning.

The steps of Feature Extraction technique are:

- Collect the data that is in the form of Image, Text, audio or any other format.
- Improve the data with removing the noise and irrelevant information that might be interfere with the feature extraction process.
- Convert the data into a format that could be processed by Feature extraction algorithms.
- Extract the relevant information from the converted data.
- Select the important features and most informative features from the extracted features for machine learning model.
- Finally create the new features by combining the exitence features in a way to improve the efficiency of machine learning model.

3.3.4 Data Argumentation

Deep learning based methods provide high accuracy in detection of chicken disease. However, Deep learning based models heavily depend upon the large data size to avoid the overfitting. In our case while detection of chicken disease we need large data set for train our model. We need large number of the features to efficiently train our model. Unfortunately, many applications don't access to big data such images.

Data Augmentation is used in that case where the data set is small or collecting the additional data set is expensive. Data Augmentation is a technique that is used to increase the data set artificially by using the exiting data set in order to avoid the overfitting. Data Augmentation technique is applied according to application. There are various functions that are used in Data Augmentation. For our project the basic functions like Horizontal flip, vertical flip, zoom, rotation etc that we are used. Following basic functions of augmentation are shown.

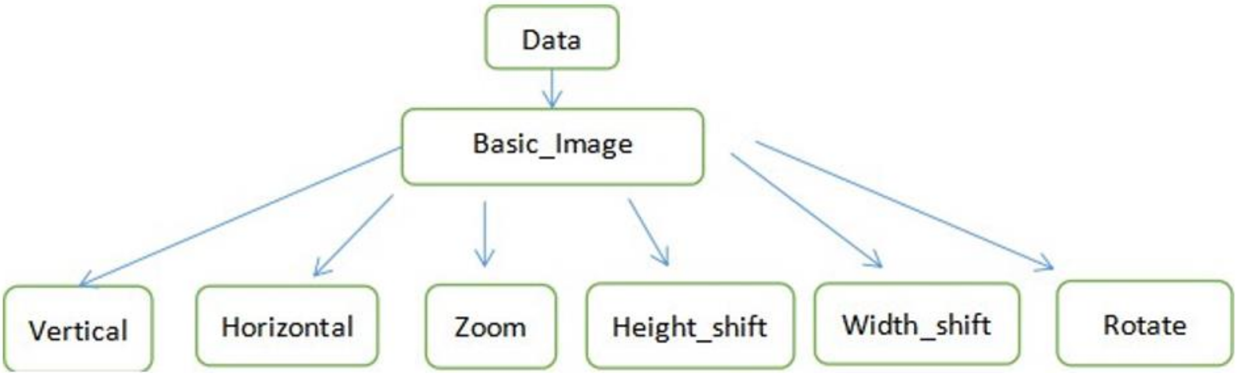


Figure25 Basic functions of Augmentation 1

These functions are implemented using the ImageDataGenerator class on the Keras Api on python interface. Keras used the ImageDataGenerator class to perform the Data Augmentation.

3.3.4.1 ImageDataGenerator

Image Data Generator is the class in Keras deep learning library that generates the batches of image data with real time augmentation. By using this class, it is efficiently feed the large amount of data into deep learning model during training. ImageDataGenerator provides the varity of options for configuring the data augmentation process zoom, vertical filp, horizontal flip and shifting.

3.3.4.2 Keras

Keras is high level language library that is written in python that provide the simple and flexible way to train the model. Keras is an user friendly library in deep learning that provides the easy way to train a model. Keras can run on different other back end engine such as Tensor flow and Theano. There are multiple application where we can used the Keras like Object detection, image classification etc. It includes multiple per-trained model. We used the MobilenetV2 in our project.

3.3.4.3 Results of Augmentation

Augmentation is required where we have different classes of image data set and all classes having random number of data set. So augmentation is applied on it that equalize the data set all classes. Small data set class require large number of Augmentation while the large data set require small number of augmentation algorithm. In short we could say that Data augmentation is used to balance the data set of classes. Amount of augmentation required is calculated through following:

$$Required = Desired \frac{Images}{total\ images}$$

We trained MobileNetv2 on UN-augmented data set and augmented dateset. The dateset used was the artificial data set.

3.3.5 Data Splitting

In above section, we cover the data augmentation in which we studied that if our data set is small and not reach our requirement or imbalance creation in our classes, the data augmentation is used to manage it and avoid the overfitting. In this section we will discuss the data splitting. Our model perform well on the trained data but when our model face the unseen data our model do not give us the value able accuracy. Overfitting is also increased on unseen data. So, data splitting is used to overcome these issues.

Data splitting is also crucial step in pre-processing. Data splitting split the data into training, validation and testing [10]. Training set is used for model development. Validation set is used for hyperparameter tuning. Hyperparameter tuning used for increasing the performance of model. Testing set is applied on the unseen data to estimates its accuracy. The purpose of the data splitting is to evaluate the performance of machine learning model accurately.

3.4 Model Training

3.4.1 Overview

In this chapter, we will discuss the CNN model and we will study about the different architects of the model that we used in our project in order to detect the chicken diseases. Model is the main part of machine learning algorithms and our project too. We used the deep learning models here that is called Deep convolutional neural network or DCNN. We could not say before this model is best fit for our project. We used the hit and trial method that is used to select the model for our project. We train the model and check the accuracy and confusion matrix. By using this information we concluded which model is best for our situation. Now we would discuss these

3.4.2 Deep CNN

Deep CNN is special type of the neural network that mainly focus on extracting the features of videos and images. It has many applications but most commonly it is used in Image processing and Computer vision. DCNN has multiple layer but common layer are:

- Convolutional layer
- Pooling layer
- Fully connected layer

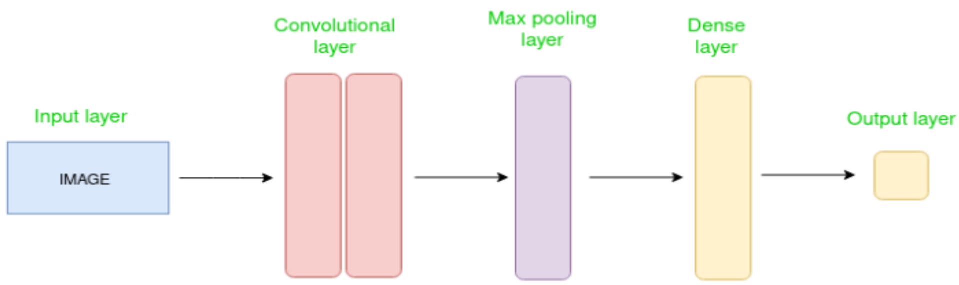


Figure26 CNN Model 1

All these layers are clearly mentioned and perform on input. These layers combine to get the multiple features from the input data. The term deep is used in this which means its having hundreds or thousands of layers in the network. The larger the layer complex scenario could be solved.

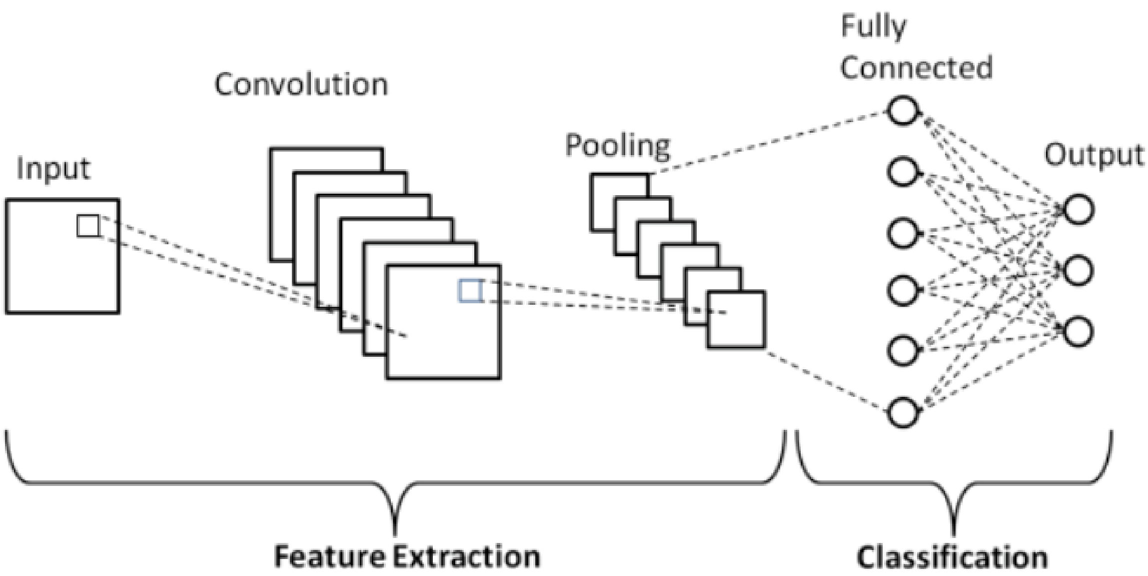


Figure27 Architect of Deep CNN models 1

3.4.2.1 Convolutional Layer

The most important layer in DCNN is Convolutional layer. It multiplied the data weight taken from the network with the input. A filter is applied during multiplication that covered the whole area from top to bottom and left to right. After the multiplication process the total inputs are summed and providing a unique value.

3.4.2.2 Pooling Layer

Pooling layer is used to extract the only most important information and reduce the size of the image by reducing the features. If the all multiplication sum are maximum numbers at the output, it is called maximum pooling. After the convolutional layer this layer is performed and these both layers are performed again and again and at the end there is fully connected layer.

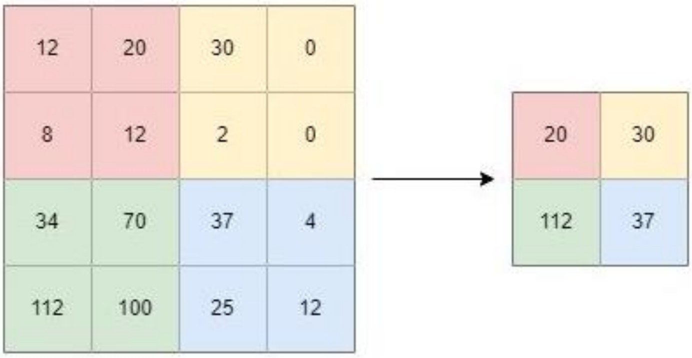


Figure28 Architect of Deep CNN models 1

3.4.2.2 Fully Connected Layer

The fully connected layer is also known as the dense layer [11]. It takes the output of convolutional and pooling layer and flattens it into a one dimensional vector.

3.4.3 MobileNetV2

MobileNetv2 is a deep convolutional neural network that was also built by Google in 2018 [12]. It is the most popular model for mobile. It has 155 layer with sequence of convolutional layer, pooling layer and fully connected layer [13]. It is light weight and efficient architecture that could perform well on the mobile devices. The architect of mobilenetv2 is given below. In figure 128 by 128 image is used. It pass through the convolutional layer, pooling and finally at fully connected layer.

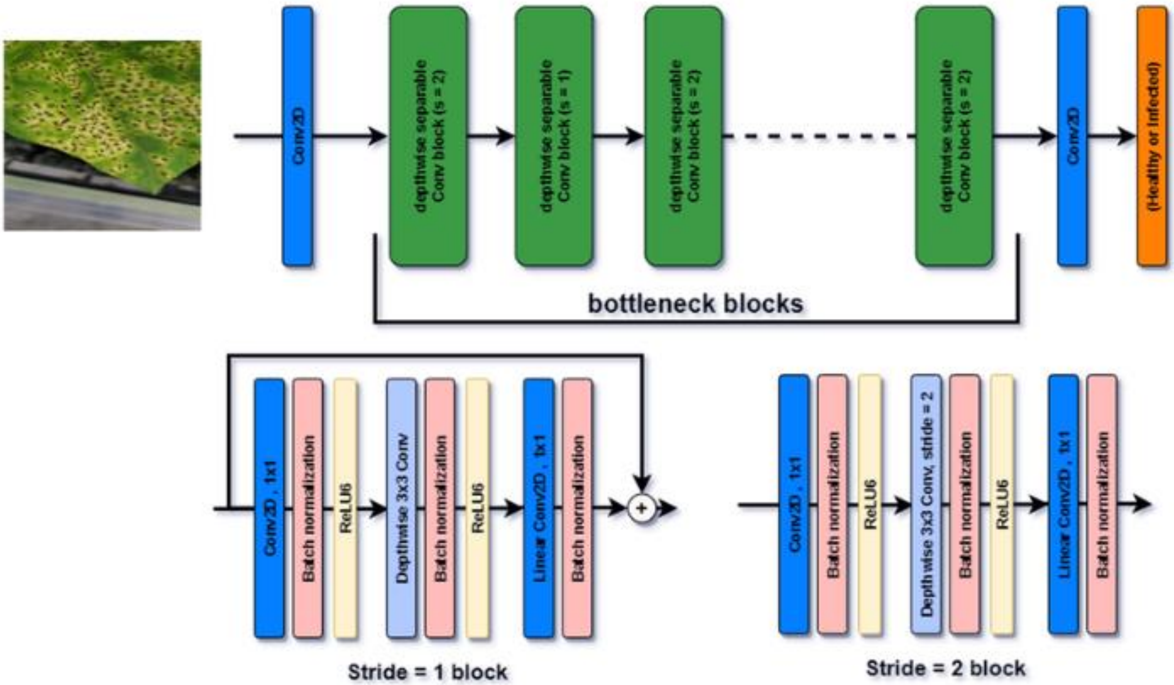


Figure29 Architect of Deep CNN models 1

In this chapter, we briefly discussed the model that are used in deep CNN and in next chapter will conclude that which one is best for our requirement.

3.5 Real Time Monitoring and Disease Detection

3.5.1 Overview

In this chapter, we will discuss about the deployment of our project. We implemented our project on platform Smart phone. If we talk about the smart phone we implemented this using mobile application on android. we will discuss how we deploy this using application on smart phone.

3.5.2 APP development

3.5.2.1 Overview

In this section, we describe the development and deployment of the smart phone application for our project, which is designed to control, monitor, manage the environmental factors of poultry farm and detect coccidiosis, Newcastle and Salmonella using machine learning. The application was built using the Flutter framework and is lightweight [14].
It has been tested on various screen sizes and resolutions and is scalable.

3.5.2.1 Purpose of APP

The purpose of the app developed for this project is real time monitoring, controlling and managing and to aid in the early detection and diagnosis of three major diseases: coccidiosis, Salmonella, Newcastle.
The main purpose of the app is to control the various factors like heat, temperature, light intensity, feed and water intake. We can monitor data from the APP or can change the status of any parameter like we can on or off the fan by clicking the button on the APP.

With the increasing prevalence of these chicken diseases, there is a pressing need for accessible and efficient screening methods that can be used poultry farms. The app aims to address this need by leveraging machine learning algorithms to analyze images captured from a camera and providing users with real-time information about their chicken health.

By utilizing advanced image processing techniques and machine learning algorithms, the app is able to analyze the images captured from the camera in real-time and provide users with accurate and reliable information about the presence of coccidiosis, Newcastle, Salmonella.

Moreover, the app is designed to be user-friendly and accessible to a wide range of individuals, regardless of their technical expertise or familiarity with chicken health. The interface is intuitive and easy to navigate, and the app provides clear and concise information about the health status of chickens at poultry farms.

In summary, the purpose of the app is to provide a reliable and accessible screening tool for the monitoring of poultry farm environmental parameters and for early detection and diagnosis of coccidiosis, Newcastle, Salmonella, ultimately improving overall chicken health outcomes [15].

3.5.2.2 APP' Features and Functionality

The smart phone application is designed to be user-friendly and accessible to individuals with little to no experience with imaging or machine learning. It provides a simple, intuitive interface that allows users to easily capture images using their smart phone camera or select an image from their photo gallery.

When a chicken image is captured or selected, the application passes the image to the machine learning model for classification. The model is a binary classifier that can classify an image as either healthy or infected with coccidiosis, Newcastle, Salmonella. The classification results are displayed to the user in real time under the image, with the image class (healthy or infected) and a confidence score represented as a percentage.

3.5.2.2.1 Home page of APP

The home page of APP indicates the University logo and shows the next direction for the further processing.

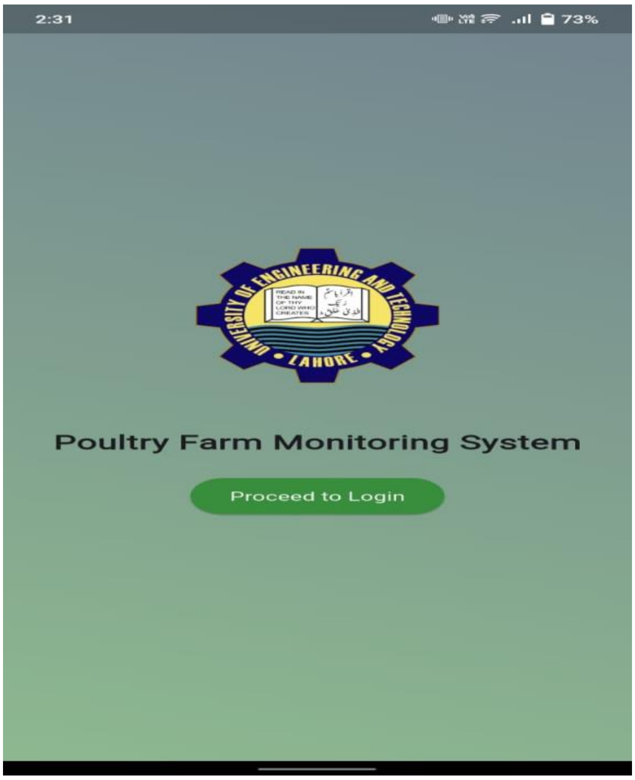


Figure30 Home page of APP 1

3.5.2.2.2 Login Page

After the home page, login page appears where we add the information about itself. As we put the username and password on the particular area, the page moves to next one. We can also proceed with next page by merely clicking the proceed to login button.

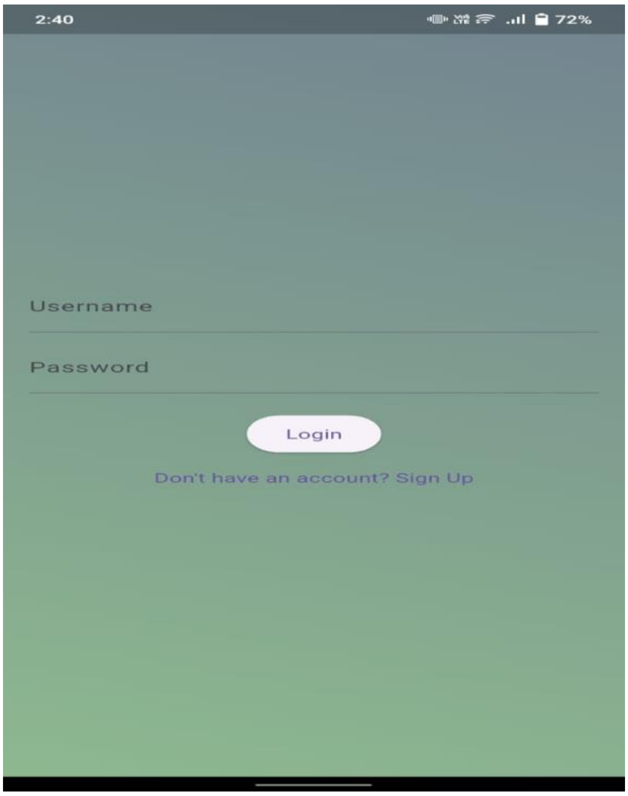


Figure31 Login Page of APP 1

3.5.2.2.2 Selection Page

At this page, we have two options about monitoring and detection. If we click on the monitoring button, then the different parameters appear on the next page and if we click detection button than several options appear on the page.

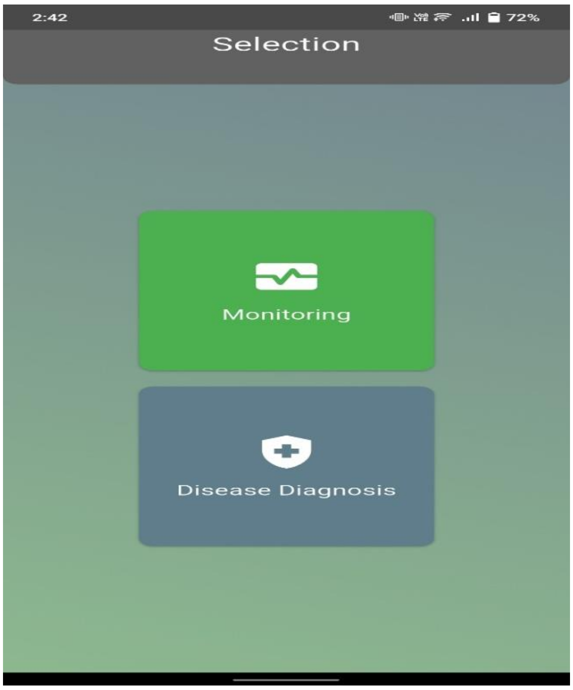


Figure32 Selection page of APP 1

3.5.2.2.2 Monitoring Page

After clicking the Monitoring button, the next page appears and shows the cycle, temperature, Humidity, Light, Water, Feed according to the environmental condition of the poultry farm.

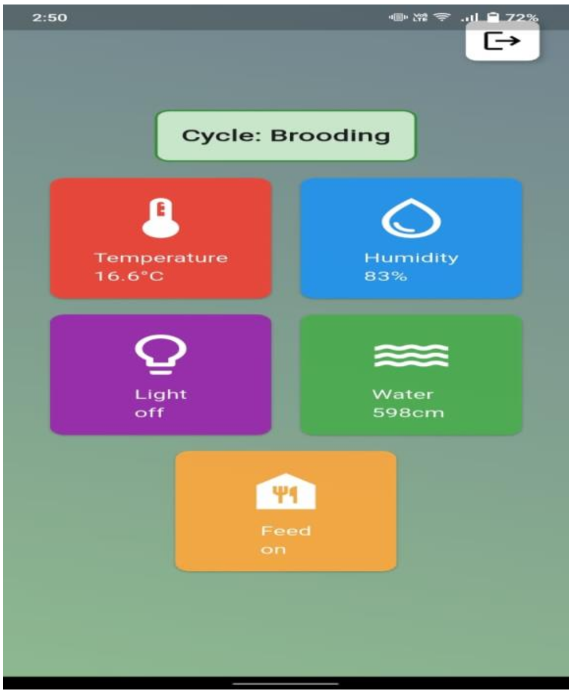


Figure33 Monitoring page of APP 1

3.5.2.2.2 Temperature Button

After clicking the Temperature button, the details about the temperature shown, as the page shows the temperature reading and condition of fan and heaters' status.

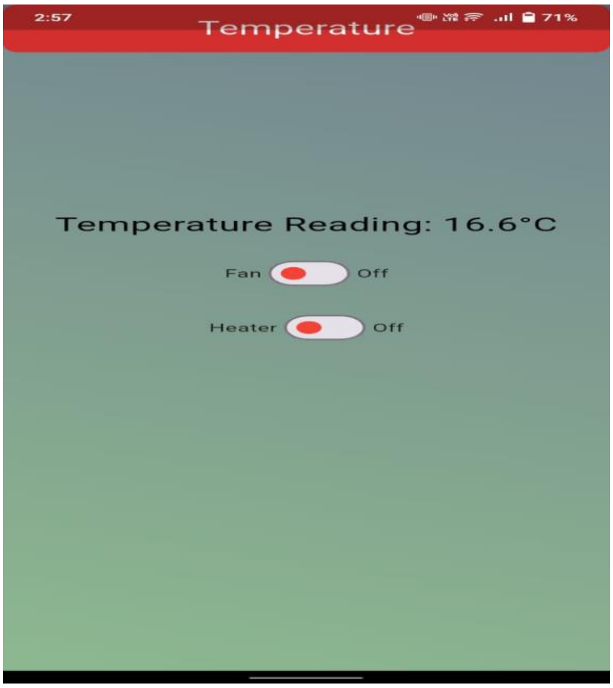


Figure34 Parameter Page 1

As by clicking any button like by clicking the temperature button as given below, the relevant information about the parameters appears on the screen. We can change the status of fan and heater by clicking the button from the app. It means we can control the parameter by clicking the button from app (manual control) and by using the information from the firebase (automatic control).

3.5.2.2.3 Disease Diagnosis

As we can select the detection of diseases of option after Login page instead of Login page.

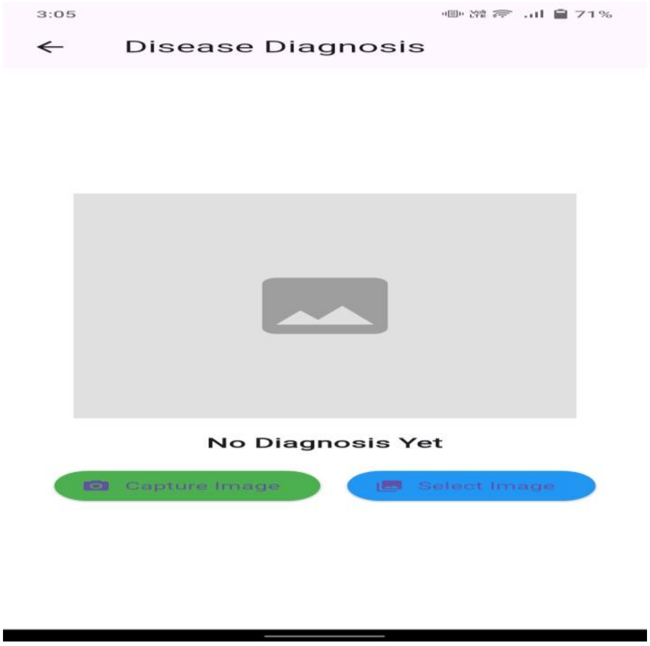


Figure35 Disease diagnosis page 1

3.5.2.4 Capture Images from Mobile Camera

The mobile app has two options about the classification of images. We can capture images from the Mobile camera for the early detection of diseases and for real time monitoring of chickens at poultry farms.

Ater clicking the Capture image button, camera of Mobile open for a purpose to take a picture.

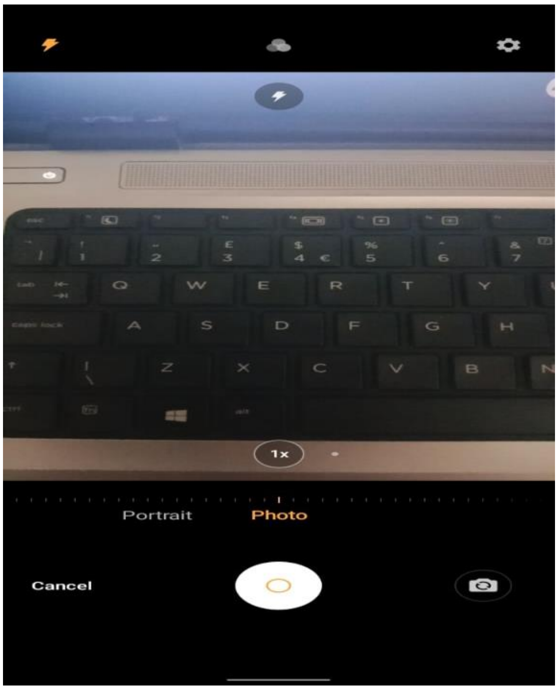


Figure36 Capture Image 1

3.5.2.3 Classify images from Gallery

The application also allows users to select images from their photo gallery for classification. To do this, the user can select the Select Photo option from the bottom navigation bar and then tap on the Start button.

This will take the user to the "Open from" page, where they can select the desired image from their smart phone storage. Once the image is selected, the application will return to the Select Photo screen and display the classification. Below show the screen from where figure selected form gallery by clicking on icon at the bottom of screen.

Overall, the smart phone application for our project provides a simple, accessible, and reliable way to detect coccidiosis, Newcastle, Salmonella using machine learning. It has been designed to be user-friendly and scalable, and has been tested extensively to ensure its performance and reliability.

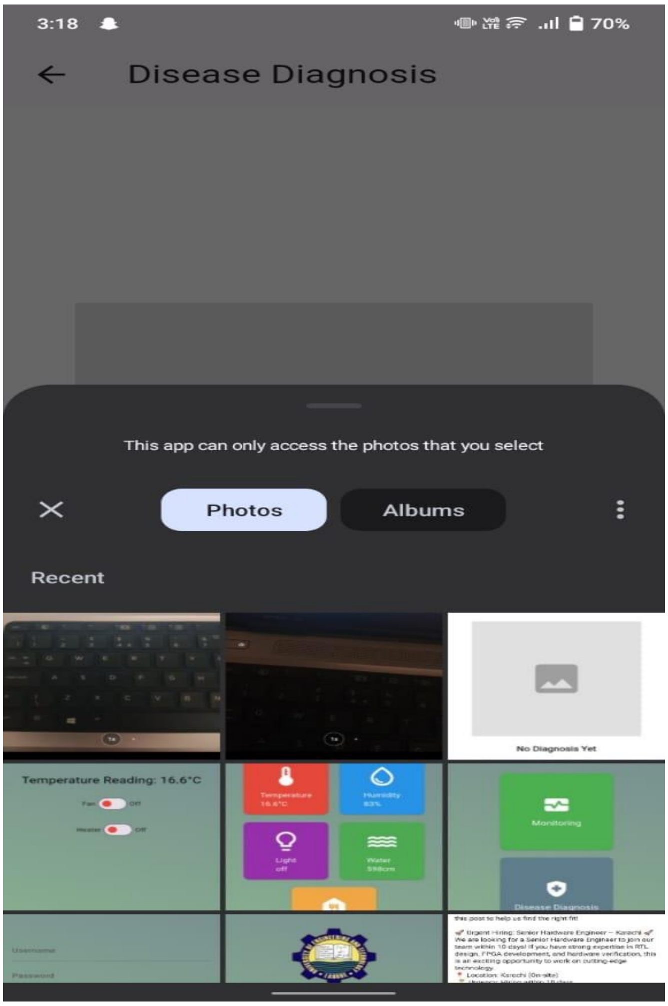


Figure37 Select Image from Gallery 1

3.5.2.4 Detection Result

The app developed in this project is capable of detecting three common chicken diseases: coccidiosis, Newcastle, Salmonella. The algorithm used in the app is trained on a large dataset of images to detect these diseases accurately. The algorithm takes the image as input and processes it through a deep learning model to identify whether the image contains any signs of these diseases.

Once the algorithm has processed the image, the app provides the user with the diagnosis of the chicken disease detected in the image. The user can then take this information to their doctor for further examination and treatment. This feature can be very useful for local areas poultry farms.

The app provides the results of the diagnosis in a clear and concise manner, making it easy for the user to understand. The results are presented on the same screen as the image and include the disease detected.

Overall, the disease detection feature of the app is a powerful tool for early detection and diagnosis of common diseases. It is user-friendly and accurate, making it a valuable resource for people who want to monitor their poultry farm's environmental conditions and want to detect early detection of diseases.

Figure38 Disease detection results 1

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Overview

In this chapter we will discuss the results of the terminologies applied to make our system successful and models that are discussed in previous chapter. We will compare these models w.r.t different situation including training time, accuracy. We will also compare these models that which one is best for Mobile application and real time detection.

4.2 Firebase interfacing with hardware

As the data of sensors are integrated by the hardware on firebase which play a crucial role in Real time monitoring

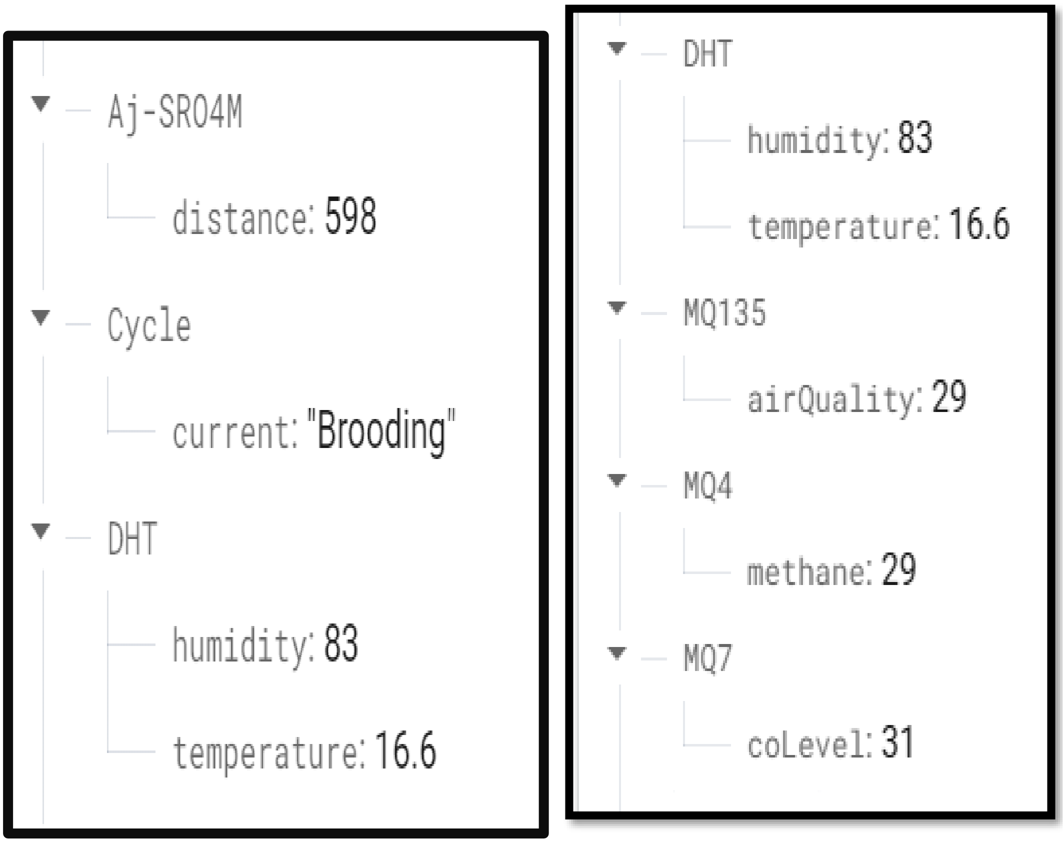


Figure 39 Interfacing with hardware 1

4.3 Firebase interfacing with Mobile APP

As the Hardware is controlled by the information provided by the firebase normally. But we add the feature in our APP to control the system directly through different options on the Mobile APP interfacing as we discussed above.

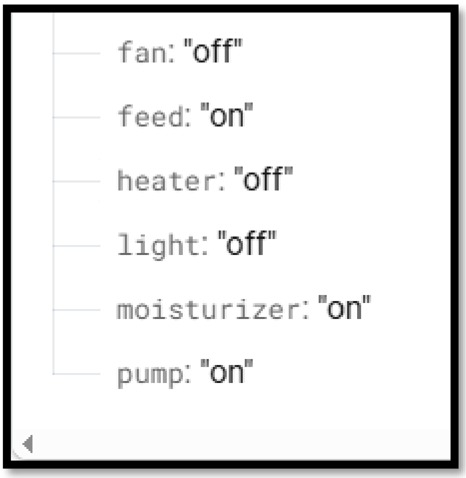


Figure40 Interfacing with Mobile APP 1

4.4 Simulation Results

As we proposed a software sample of our system to test the working condition of our proposed model.

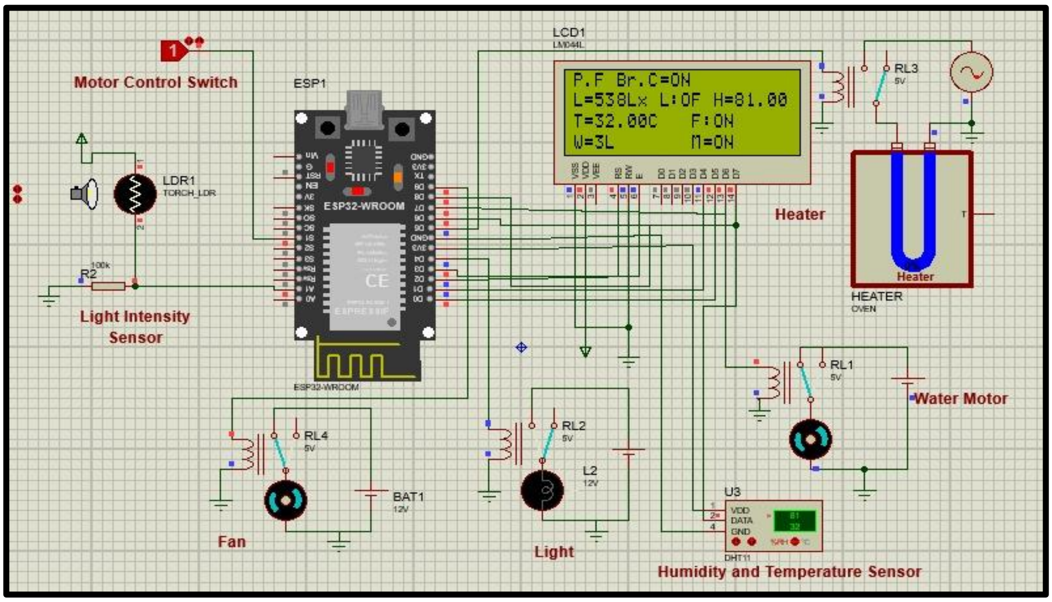


Figure41 Simulation of system 1

The given simulation indicates the best performance of our real time monitoring of the data. For example, if the temperature of the system decreases the heater turns on and if the temperature increases the fan turns on and heater turns off wisely.

The motor control switch indicates the water motor control by which we can on and off the motor water pump. The light intensity sensor indicates the light control for our system [16]. Humidity and temperature sensor indicates the control of humidity and temperature of our poultry farm. The LED indicates the status of water pump, moisture pump, intensity of light, temperature of the system indicated by the sensors.

4.5 Pre-trained Model performance

Following Table shows the performance of the pre-trained model.

loss	accuracy	val-loss	Val- accuracy
0.3109	0.8897	0.1927	0.992
0.1455	0.9492	0.1452	0.9537
0.1116	0.9601	0.1240	0.9595
0.0910	0.9692	0.1156	0.9578
0.0773	0.9758	0.01047	0.9657

4.5.1 MobileNetV2 Results

We select mobilenetv2 poultry diseases diagnosis because the architecture of MobileNetV2 is designed to be efficient, lightweight, and optimized for mobile and embedded devices while maintaining high performance.

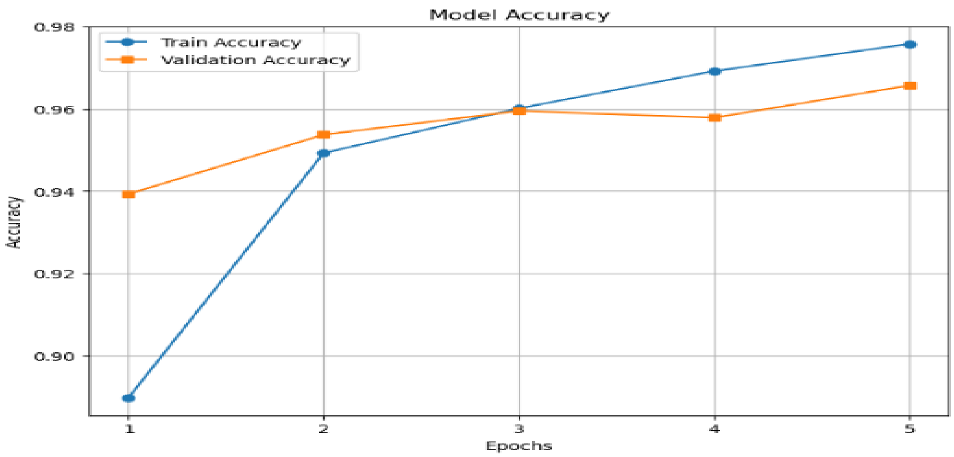


Figure42 Performance Graph 1

4.5.1.1 Training process and performance

We trained a MobileNetV2 model using transfer learning on a dataset of chicken disease images, with the goal of accurately classifying images into one of four classes: class A, class B, class C, or class D. The dataset consisted of pre-processed images of size 128x128 pixels, with a total of 1000 images in each class almost. We used an ImageDataGenerator object for data augmentation, which included rescaling, validation splitting, random rotations, shifts, shears, zooms, and flips.

The model performed exceptionally well overall, with excellent accuracy and precision which shows that the majority of occurrences were properly identified. It's important to note that the model incorrectly categorized four occurrences from the fourth class as belonging to the second class, suggesting that this is a potential subject for further research or model improvement.

CHAPTER 5: CONCLUSION AND FUTURE RECOMMENDATIONS

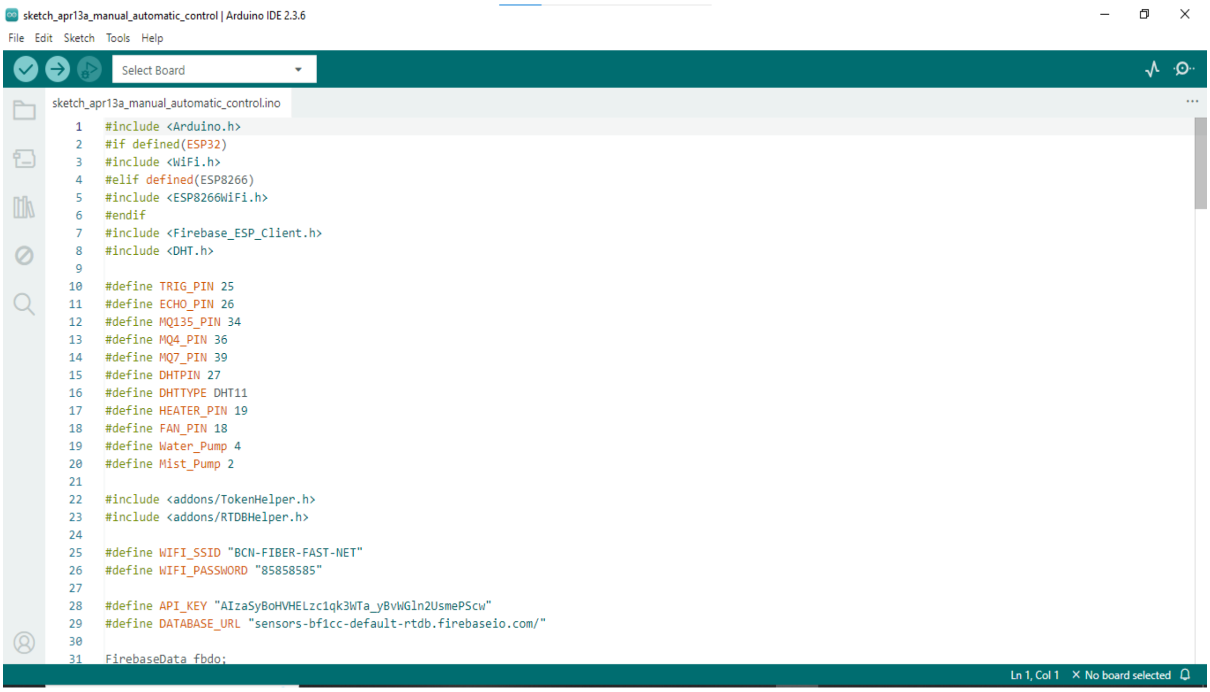
Cocciodiosis, Newcastle, Salmonella are the most common chicken diseases affecting millions of poultry farm worldwide. These diseases can lead to disaster if not detected and treated early. There are lots of poultry farms in the world that are affected by these diseases and manual method is not efficient idea to diagnose these diseases. Because manual method requires more attention that waste out time, money and efforts. Manual method also not available in all areas which creates the delay in diagnosing and treatment. Since last decade, the machine learning models play very key role in diagnosing the chicken diseases. By machine learning models especially deep learning techniques the diagnosing process is more fast and efficient. Machine learning models also cost effective and scalability than the manual method.

In this thesis, we presented a technique which is more reliable than others that are represented in literature. We trained the machine learning model by using this technique on artificially generated dataset. By using Augmentation we increased our dataset. We collected the dataset from different online platforms and also from poultry farms. We trained our model on architecture like MobileNetv2. We concluded that MobileNetv2 is best for our requirement and gave highest accuracy. We analyzed the accuracy on different factors using f1 score, accuracy score and confusion matrix. Model also compared with the models that are used in supervised leaning. We concluded that the deep learning model MobilenetV2 is good for the patients in sense of time, accuracy and early diagnosing.

We deployed our machine learning model on Linux and Mobile app. Mobilenetv2 is also best for app deployment. We used the live capture image and show the results while on app. We used the storage picture. we used the Web-cam to take the live picture.

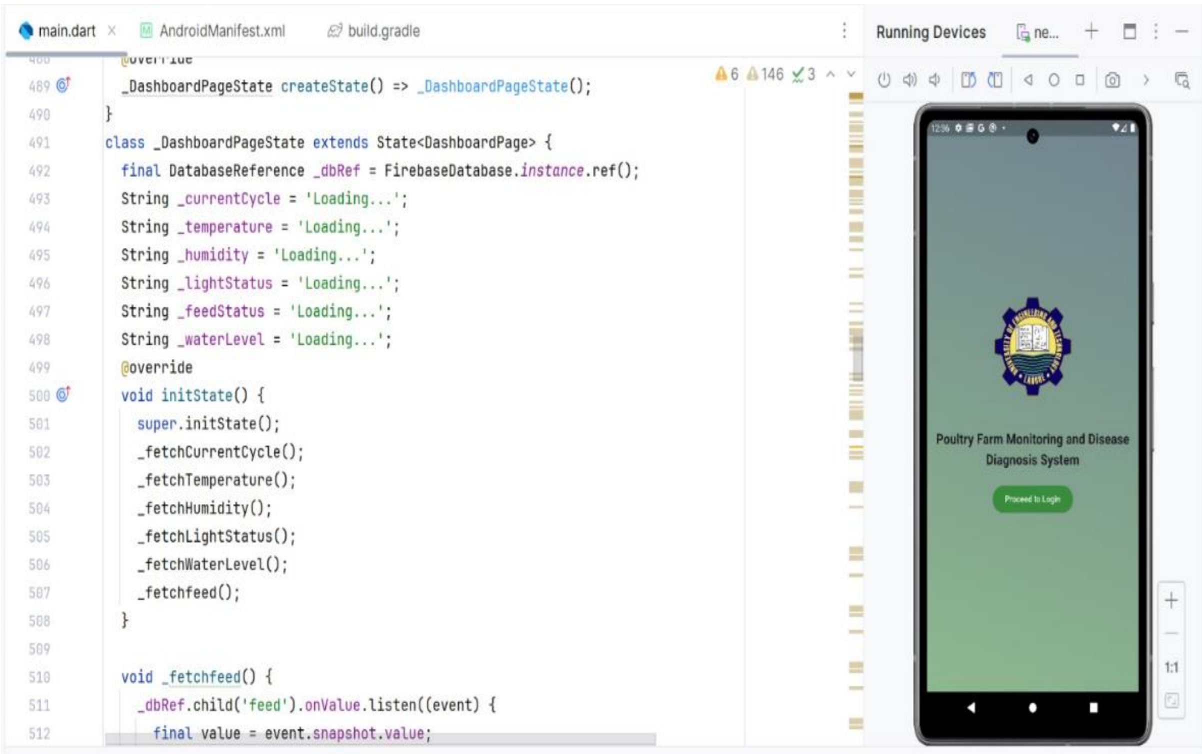
Appendix

Code for ESP32:



```
#include <Arduino.h>
#if defined(ESP32)
#include <WiFi.h>
#elif defined(ESP8266)
#include <ESP8266WiFi.h>
#endif
#include <Firebase_ESP_Client.h>
#include <DHT.h>

#define TRIG_PIN 25
#define ECHO_PIN 26
#define MQ135_PIN 34
#define MQ4_PIN 36
#define MQ7_PIN 39
#define DHTPIN 27
#define DHTTYPE DHT11
#define HEATER_PIN 19
#define FAN_PIN 18
#define Water_Pump 4
#define Mist_Pump 2

#include <addons/TokenHelper.h>
#include <addons/RTDBHelper.h>

#define WIFI_SSID "BCN-FIBER-FAST-NET"
#define WIFI_PASSWORD "85858585"

#define API_KEY "AIzaSyBoHVHELzc1qk3WTa_yBvWGln2UsmePScw"
#define DATABASE_URL "sensors-bf1cc-default-rtdb.firebaseio.com/"

FirebaseData fbdo;
```

Code for APP:



```
  _DashboardPageState createState() => _DashboardPageState();
}
class _DashboardPageState extends State<DashboardPage> {
  final DatabaseReference _dbRef = FirebaseDatabase.instance.ref();
  String _currentCycle = 'Loading...';
  String _temperature = 'Loading...';
  String _humidity = 'Loading...';
  String _lightStatus = 'Loading...';
  String _feedStatus = 'Loading...';
  String _waterLevel = 'Loading...';
  @override
  void initState() {
    super.initState();
    _fetchCurrentCycle();
    _fetchTemperature();
    _fetchHumidity();
    _fetchLightStatus();
    _fetchWaterLevel();
    _fetchfeed();
  }

  void _fetchfeed() {
    _dbRef.child('feed').onValue.listen((event) {
      final value = event.snapshot.value;
```

Code of Deep Learning:

```python
# Allow truncated images to be loaded instead of causing an error
ImageFile.LOAD_TRUNCATED_IMAGES = True

# Training model
# Make sure checkpoint_path is correctly saved
history = model.fit(
    train_dataset,
    epochs=5,
    validation_data=valid_dataset,
    callbacks=[cp_callback]
)
```

Python

```
Epoch 1/5
170/170 [==============================] - ETA: 0s - loss: 0.3109 - accuracy: 0.8897
Epoch 1: saving model to C:\Users\Muhammad Salman/deep_learning_checkpoints\mobilenetv2_cp.weights.h5
170/170 [==============================] - 2215s 13s/step - loss: 0.3109 - accuracy: 0.8897 - val_loss: 0.1927 - val_a
Epoch 2/5
170/170 [==============================] - ETA: 0s - loss: 0.1456 - accuracy: 0.9493
Epoch 2: saving model to C:\Users\Muhammad Salman/deep_learning_checkpoints\mobilenetv2_cp.weights.h5
170/170 [==============================] - 2110s 12s/step - loss: 0.1456 - accuracy: 0.9493 - val_loss: 0.1453 - val_a
Epoch 3/5
170/170 [==============================] - ETA: 0s - loss: 0.1116 - accuracy: 0.9601
Epoch 3: saving model to C:\Users\Muhammad Salman/deep_learning_checkpoints\mobilenetv2_cp.weights.h5
170/170 [==============================] - 2052s 12s/step - loss: 0.1116 - accuracy: 0.9601 - val_loss: 0.1240 - val_a
```

```python
# we decide default weigh for mode
mobv2_model = tf.keras.applications.MobileNetV2(input_shape=(128,128,3),
                                                include_top=False, # Transfer learning
                                                weights="imagenet")
```

[8] Python

```python
# model summary
# Total params: 2,257,984
# Trainable params: 2,223,872
# Non-trainable params: 34,112
mobv2_model.summary()
```

[9] Python

```
... Model: "mobilenetv2_1.00_128"
```

Layer (type)	Output Shape	Param #	Connected to
input_1 (InputLayer)	[(None, 128, 128, 3)]	0	[]
Conv1 (Conv2D)	(None, 64, 64, 32)	864	['input_1[0][0]']
bn_Conv1 (BatchNormalization)	(None, 64, 64, 32)	128	['Conv1[0][0]']
Conv1_relu (ReLU)	(None, 64, 64, 32)	0	['bn_Conv1[0][0]']

```python
import matplotlib.pyplot as plt

def plot_graph(history):
    # Ensure history contains accuracy data
    if 'accuracy' not in history.history or 'val_accuracy' not in history.history:
        print("Error: Accuracy data not found in history.")
        return

    epochs_range = range(1, len(history.history['accuracy']) + 1)  # Generate epoch numbers

    plt.figure(figsize=(8, 6))

    # Plot training & validation accuracy
    plt.plot(epochs_range, history.history['accuracy'], label='Train Accuracy', marker='o')
    plt.plot(epochs_range, history.history['val_accuracy'], label='Validation Accuracy', marker='s')

    plt.title('Model Accuracy')
    plt.xlabel('Epochs')
    plt.ylabel('Accuracy')
    plt.xticks(epochs_range)  # Ensure epoch numbers are displayed correctly
    plt.legend()
    plt.grid(True)

    # Show the graph
    plt.show()
```

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Complex Engineering Problem for FYDP

Smart Poultry Farm Management and Disease Diagnosis

CEP Statement:

Smart Poultry Farm Management and Disease Diagnosis

Brief Project Description:

This project aims to develop an automated system for managing a poultry farm efficiently while diagnosing common poultry diseases using IoT, AI, and sensor-based monitoring. The features of the project include real-time monitoring. Sensors measure temperature and humidity in the poultry environment. Automated alerts are generated if conditions exceed safe limits. For Disease Detection Using AI, Cameras and image processing detect symptoms of diseases. Mobile app interface for controlling ventilation, temperature, and lighting.

Project is proposed that has diverse groups of stakeholders with widely varying needs. The stakeholders involve farms, technologists, solution providers and business communities.

CEP Attributes:

1. **Depth of knowledge required**

Research-based knowledge of integrating contemporary devices and systems for reliable operation of the system.

Knowledge Profile: WK4 Engineering Specialization

The knowledge of Engineering specialization of sensor working and interfacing is necessary.

2. **Extent of stakeholder involvement and level of conflicting requirements**

A solution is proposed that has diverse groups of stakeholders with widely varying needs. The stakeholders involve farms, technologists, solution providers and business communities.

Knowledge Profile: WK7 Engineering in Society

Role of engineers in a society to public safety including the impact of an engineering activity i.e. economic, social, cultural, environmental and sustainability.

3. **Consequences**

This project can have significant consequences on the poultry farming business in a range of contexts including farms, technologists, solution providers and business communities.

4. **Interdependence**

The project is targeting high-level problems of farm automation which includes many component parts to ensure reliability. The complete solution requires interfacing of different devices.

Sustainable Development Goals

SDG 2: Zero Hunger

Meat yield can be improved by increasing the yield of the farms by better management using this project. More food can be produced.

SDG 9: Industry, Innovation and Infrastructure

The project has diverse groups of stakeholders involving food processing Industry, farms, technologists, solution providers and business communities. Sustainable industrialization can be achieved through connecting the farms and industry.

FYP's Societal Impact:

The users of this project are farms, food processing Industry, technologists and business communities. Sustainable industrialization can be achieved through connecting the farms and industry.

ab

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